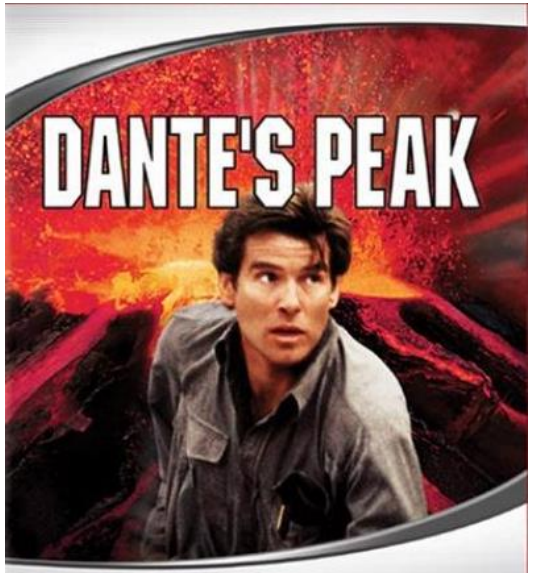
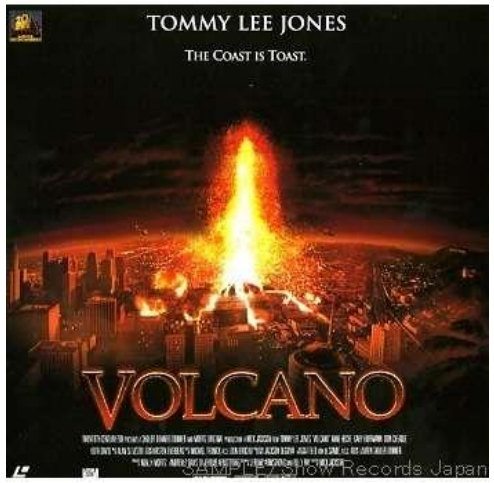
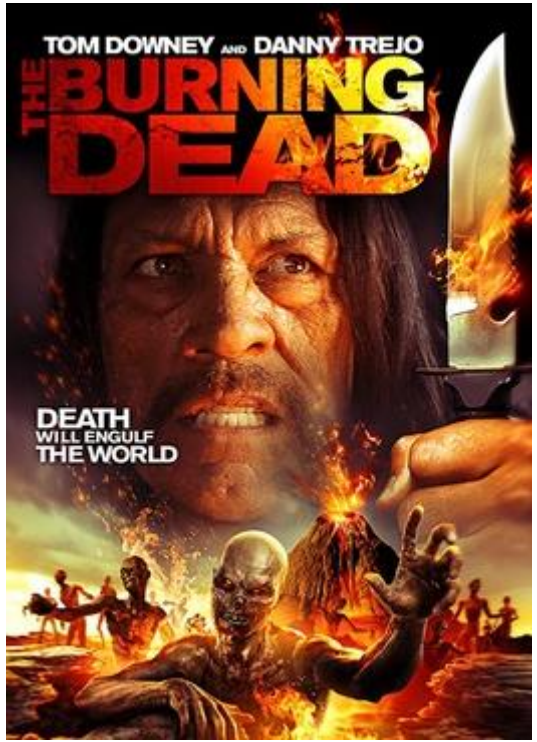
Your *Mystery Volcano* Project
Generic Instructions















Gather information

- From your homework and exercises you should know/have:
  - What volcano you're working on
  - The eruptive history of the volcano
  - The history of lahars (how big? how often? what triggers them?)
  - Any existing geological maps
  - Any existing hazard maps
  - If there is a reason to update the hazard map (different scenario? different audience?)
  - What scenario or scenarios you will focus on (what type of lahars? what volume of lahars? what size eruption, if any?)
  - A DEM of your volcano
  - Some Google Earth images of your volcano
  - A population map if available
  - A topographic or other map that shows roads, rivers, lakes, villages, cities, etc.
  - Who is your target audience (General public? Mining company or transportation agency? Tourists? Emergency managers? Land-use planners?)

Google Earth exploration

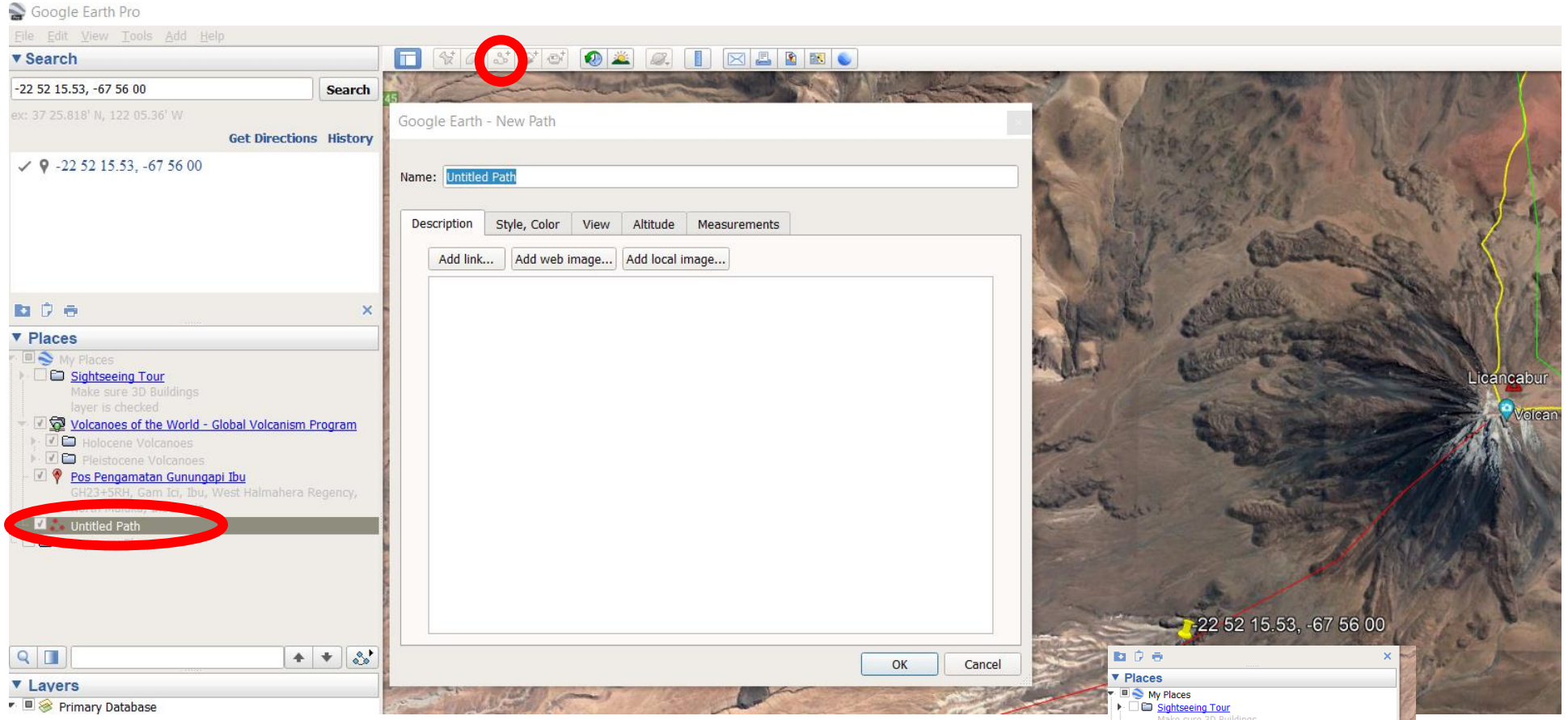
- You need to do some investigating in Google Earth
- Search for your volcano and collect some information
- You'll need the summit coordinates
- Is your summit nice and pointy? Or complex with multiple peaks or a deep crater?
- You need to make some slope profiles to help decide on an H/L ratio (see next slides as a reminder)
- Look for evidence of pyroclastic flow or other deposit fans and calculate their horizontal distance, change in height (subtract the elevation at the end of the deposit from the elevation of the summit), and calculate the H/L ratio (Look at the streams and cones presentation for a refresh)

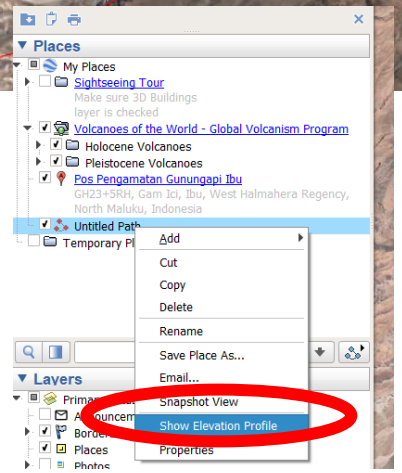
Okay, so how do we choose the H/L value?

PDC deposit fans are useful for choosing an ΔH/L cone/proximal hazard zone

- **Visualize volcano in 3-D:**
  - Find photos (Google image search, GVP)
  - Use Google Earth to get oblique views, fly around, and make a topographic profile.
- **Examine volcano records of past events:**
  - geologic maps; observatory logs; GVP info
- **Look at the geomorphology:**
  - landforms, farming, land use
- **Look for PDC units in the field**
- **Examine satellite imagery**
- **Use global record of analog volcanoes (PDC H/L values)**
- **Start with H/L of 0.3 and refine**

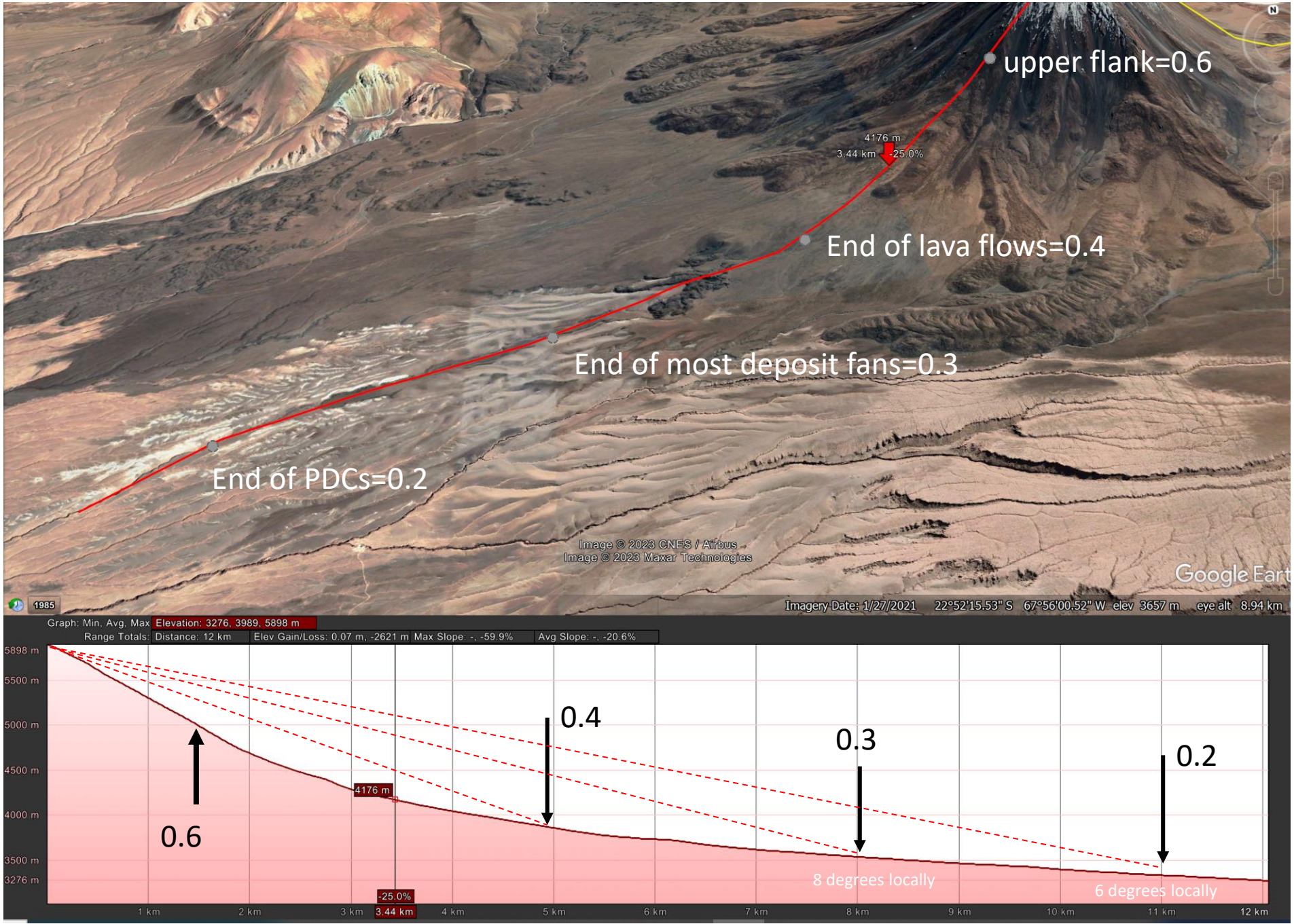
Visualize volcano in 3-D





- Click on the “Add path” button
- Click on the map to make a path
- Name and save the path
- Right click on the path name under “Places”
- Choose “show elevation profile”
- Be careful: the slope values given in Google Earth are local, not H/L from the summit

Visualize volcano in 3-D



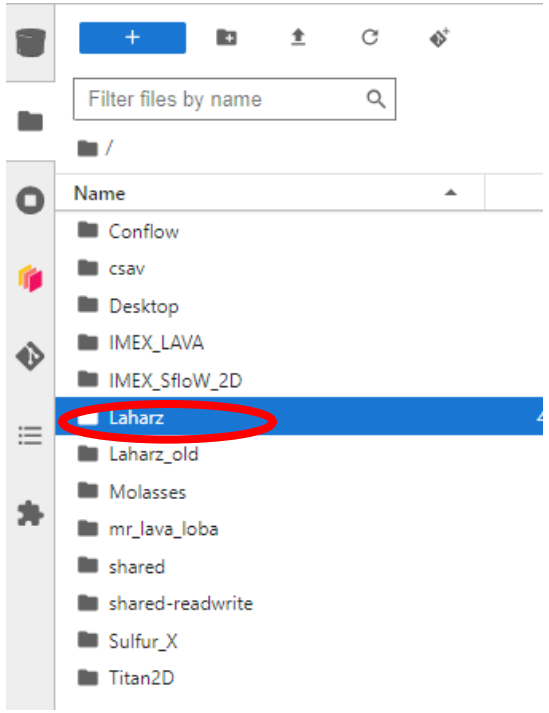
Special Cases: not a nice pointy volcano?
Add elevation in LAHARZ



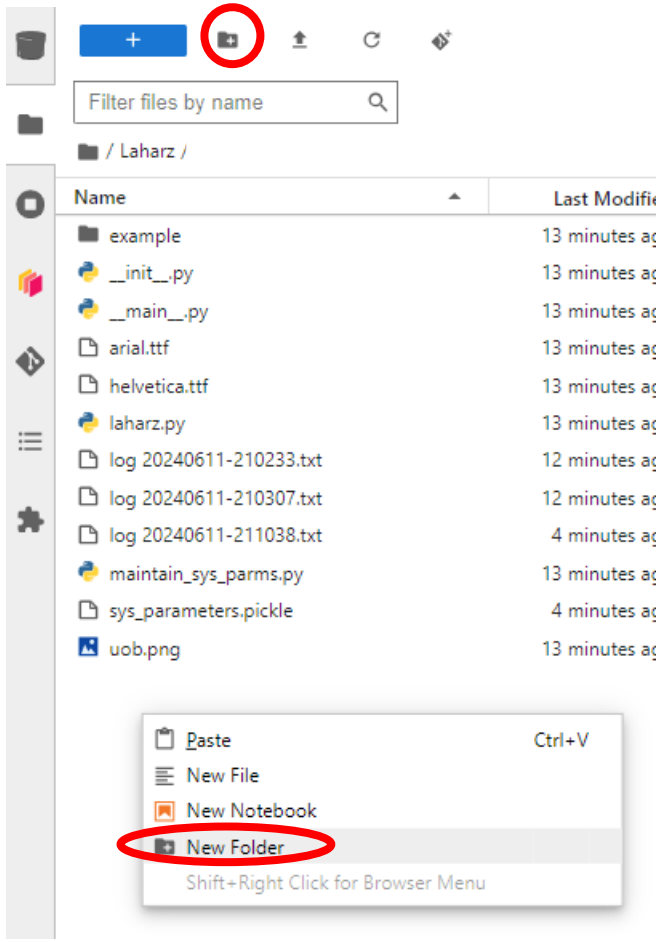
Gather more information and write it down

- What are the coordinates of your summit (search in Google Earth):
  - Latitude:
  - Longitude:
- What volumes will you model:
- What stream network thresholds will you try:
- What H/L ratio(s) will you try:
- Is your volcano pointy or complex:
  - If your volcano is complex, how much elevation do you think you need to add? (check a side profile in Google Earth):

Let’s make a folder to work in!

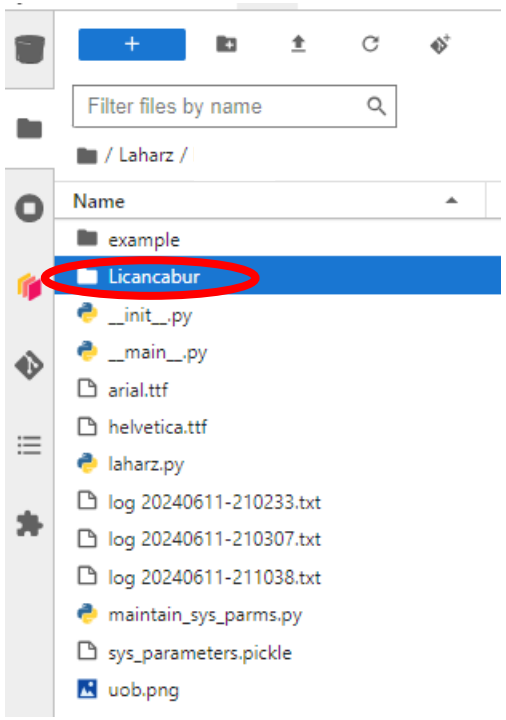


Double click on “Laharz”



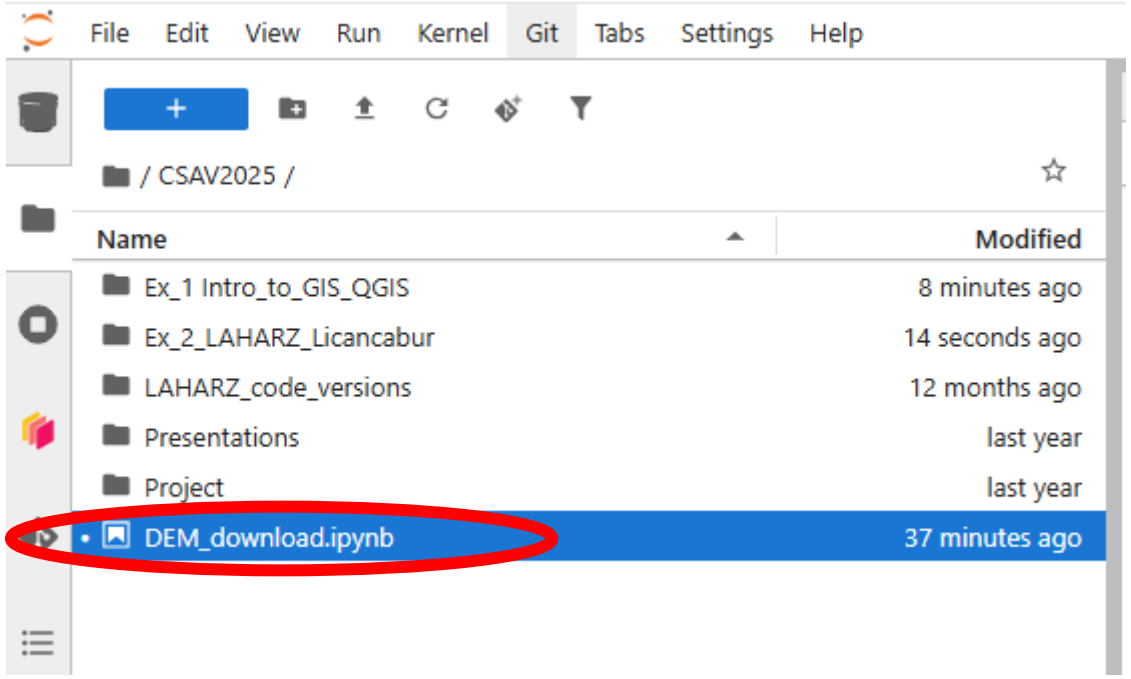
Right click in a blank space and choose “New Folder”

Or click the “New folder“ icon at the top



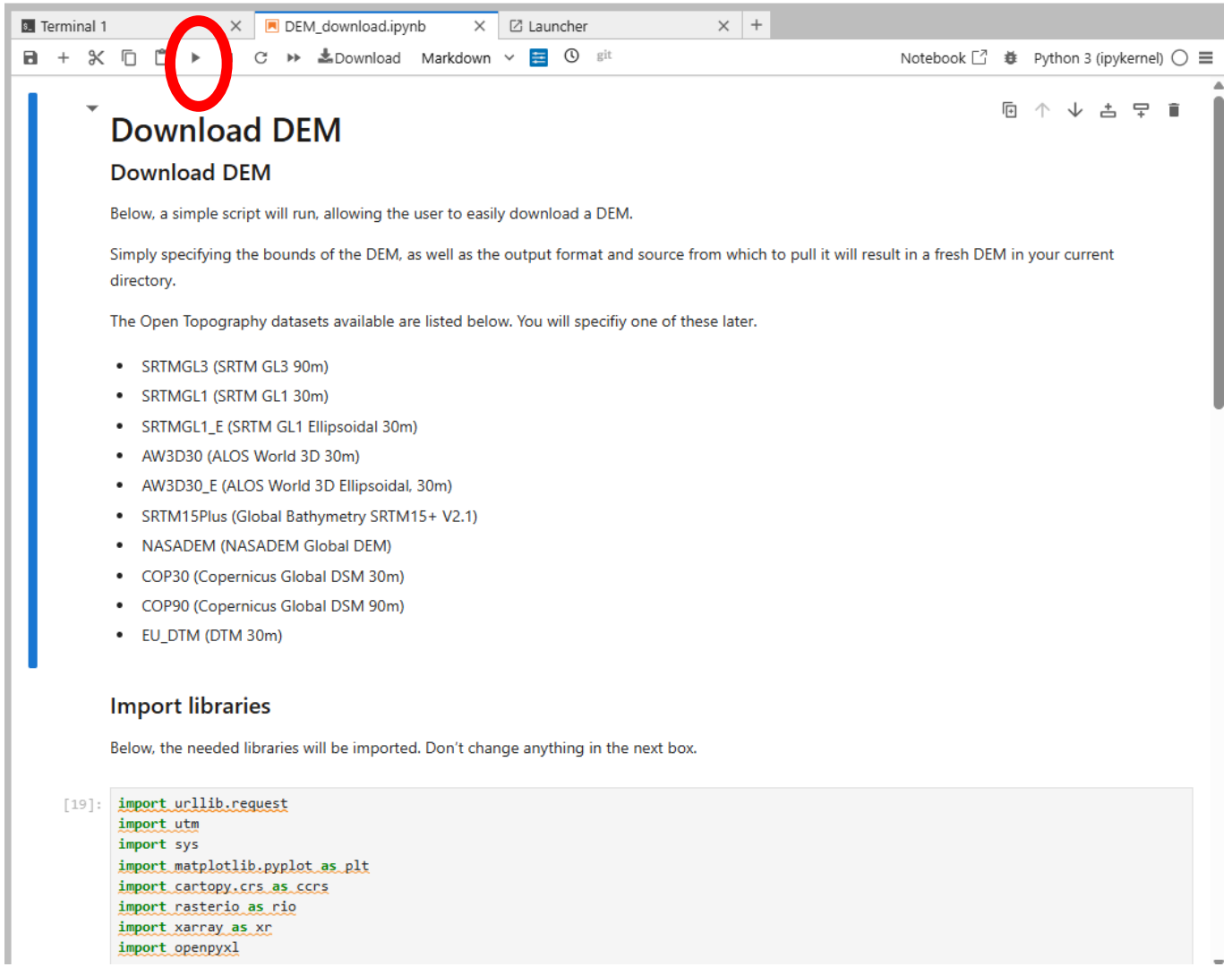
Name the new folder with your volcano name

Using VICTOR to get a DEM



- Inside VICTOR, you should have a Jupiter notebook called “DEM_download.ipynb”
- Open it

- To use the notebook, you will follow the directions in the notebook and put your cursor in the first box and click the run button at the top



Import libraries

Below, the needed libraries will be imported. Don't change anything in the next box.

[19]:
```
import urllib.request
import utm
import sys
import matplotlib.pyplot as plt
import cartopy.crs as ccrs
import rasterio as rio
import xarray as xr
import openpyxl

sys.path.insert(0,"/home/jovyan/shared/Libraries")
import victor
```

- For some boxes, you don't change any parameters, you just run it!

Specify parameters

In the box below, edit the values to specify the extent of the DEM and the output filetype.

[20]:
```
#Edit the latitude and longitude of the DEM extent in decimal degrees
north = 46.375

south = 46.0863

west = -122.7584

east = -121.9504

#Select a DEM format: Input "ascii" for ASCII format (.asc) and "tiff" for geotiff format
outputFormat = "ascii"

#Choose the Open Topography data type you want to download

dataset = "SRTMGL1"
```

- In this box, you will see there are places to specify the DEM coordinates and the file type
- Add the coordinates you need for your volcano

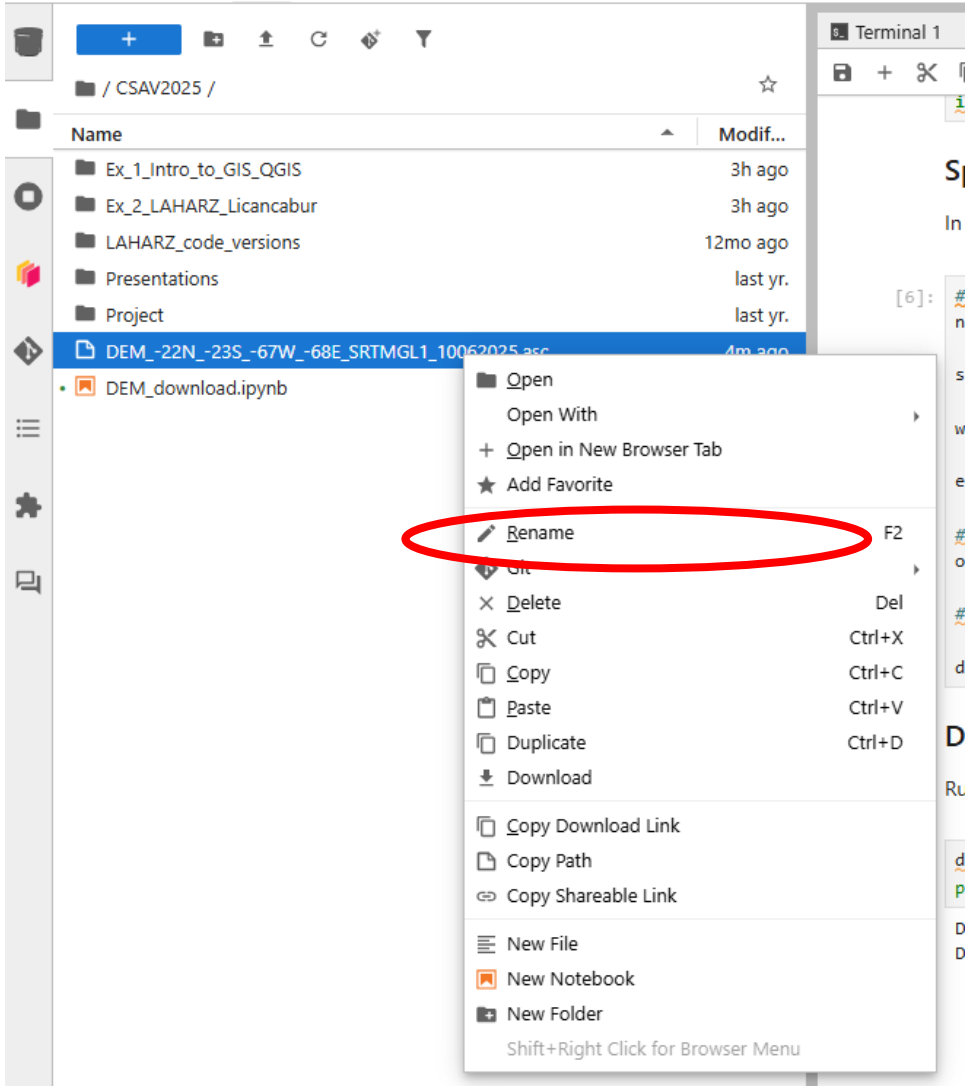
Download DEM

Run the box below without changing anything. A DEM will be downloaded into your workspace.

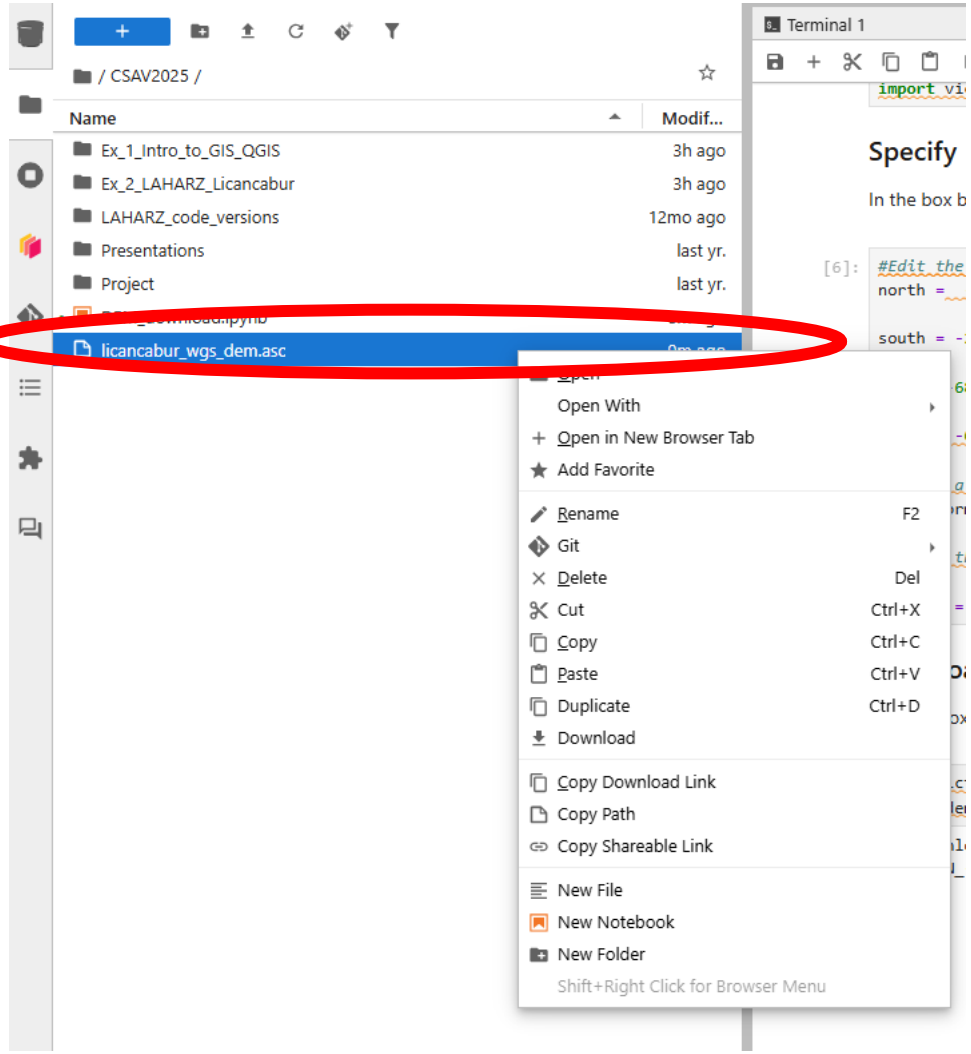
```
[*]: dem = victor.download_dem(north, south, east, west, outputFormat, dataset)
     print (dem)
```

- When you run the “Download DEM” code, you will see that it takes some time to run the code.
- You can tell because there is a [*] in the brackets
- When the * changes to a number, you should see a file downloaded!

Rename the DEM

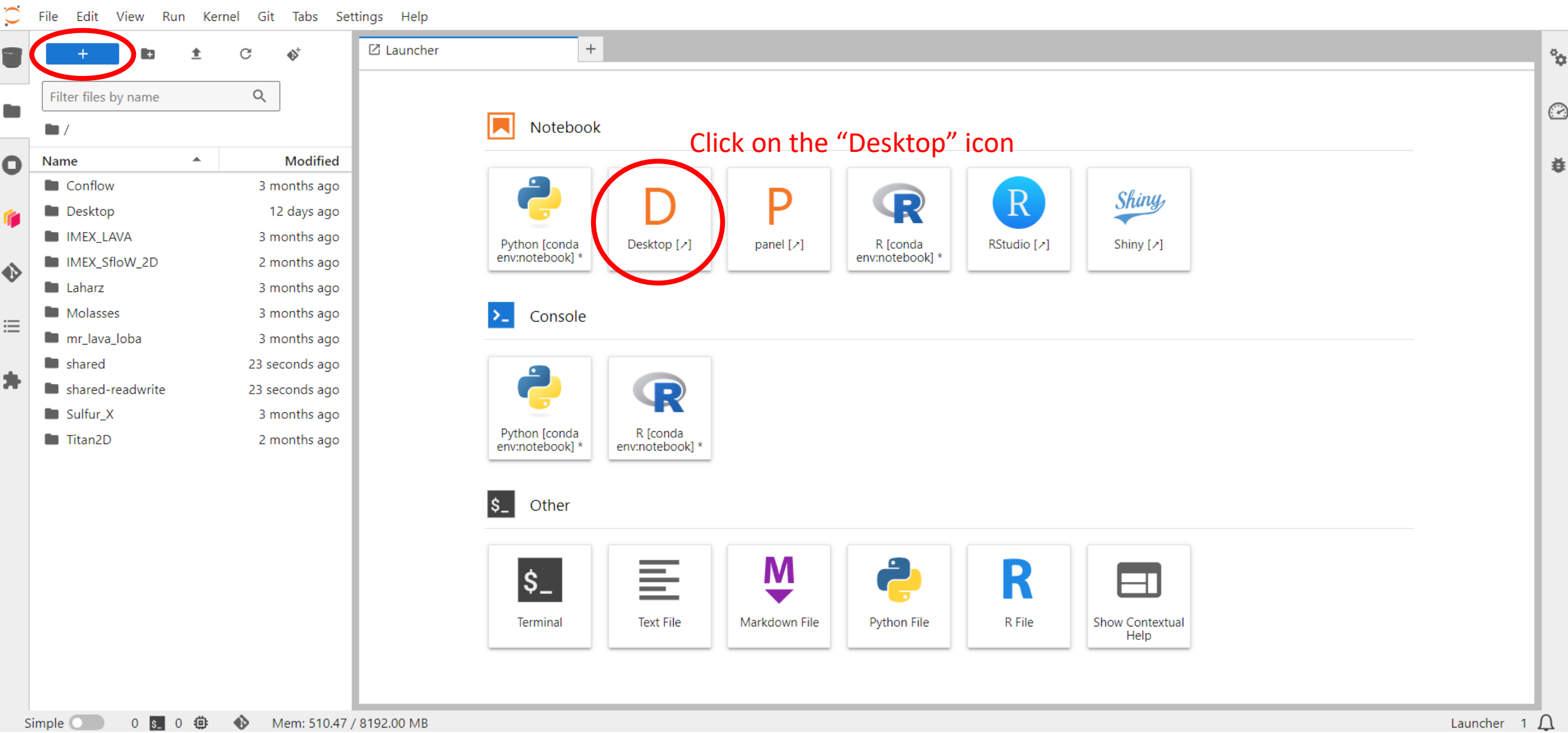


- The downloaded DEM should now be in your CSAV2025 folder
- Rename it by right clicking and choosing “Rename”
- Name it “my_volcano_dem.asc”



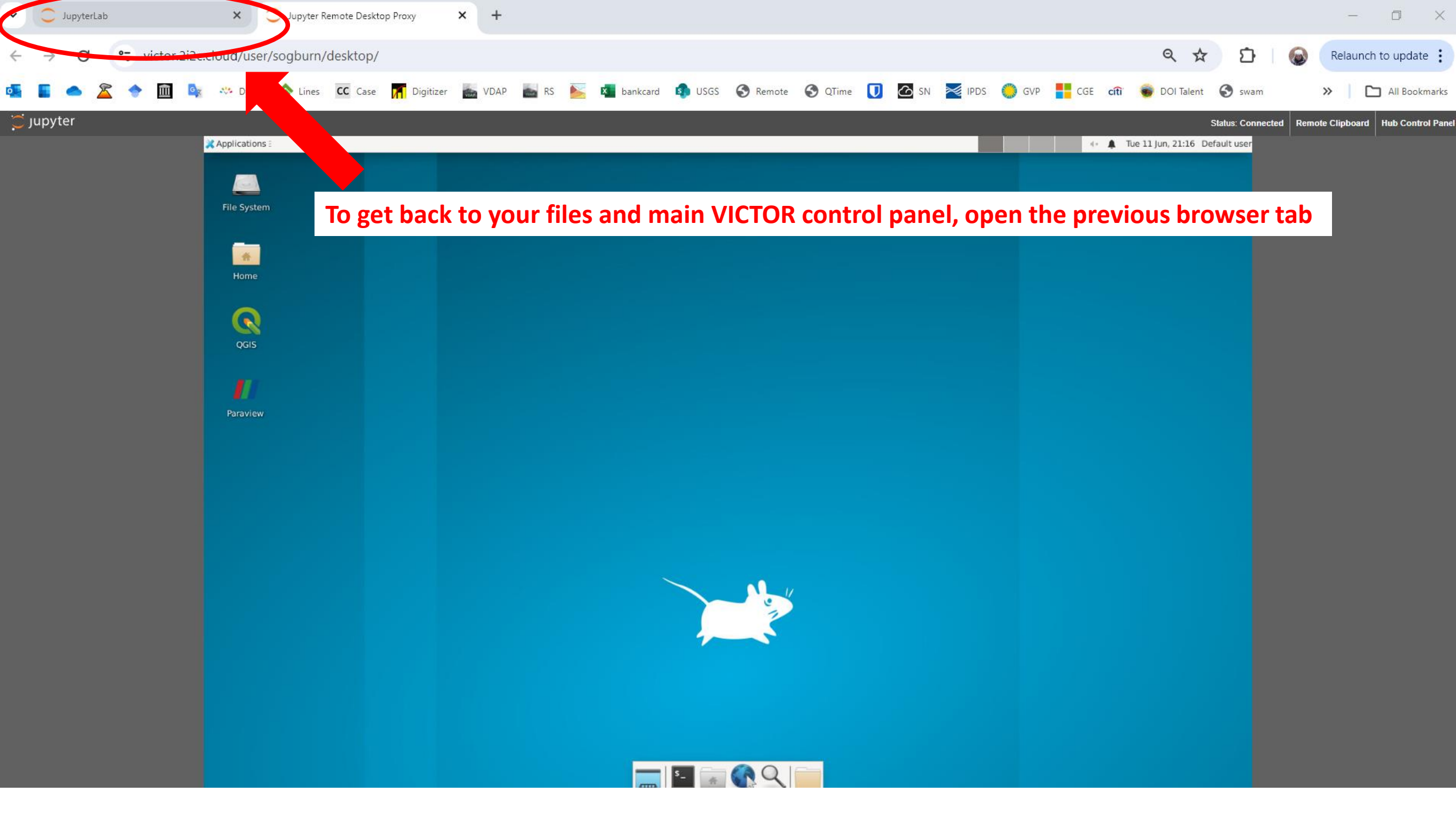
- Copy the new DEM to the Laharz/my_volcano folder

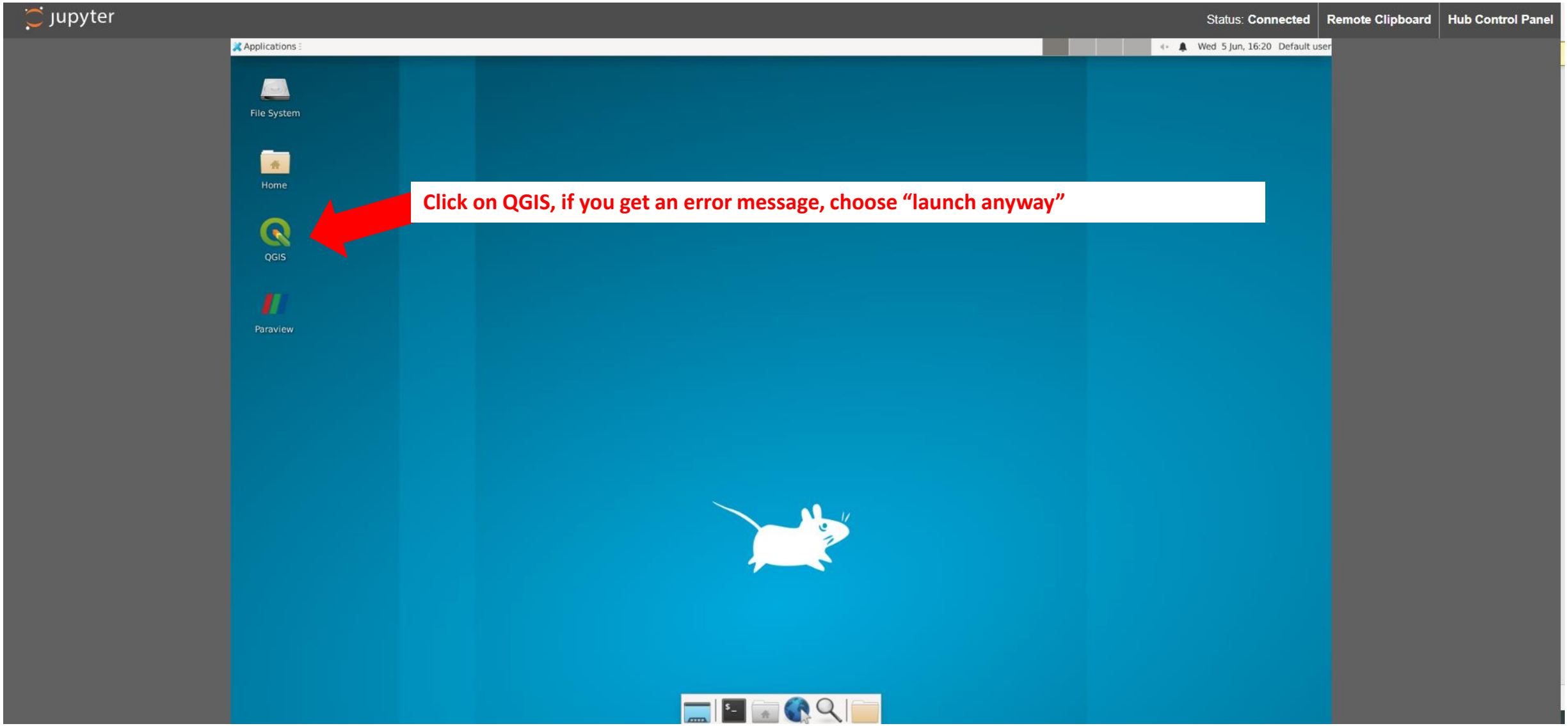
Use QGIS to make a hillshade and a filled DEM



Click on the "Desktop" icon

Tip: If you lose your "Launcher" screen, click the "plus" icon at the top left for "New Launcher"


JupyterLab
Jupyter Remote Desktop Proxy
victor.2i2c.cloud/user/sogburn/desktop/
Relaunch to update
Lines
Case
Digitizer
VDAP
RS
bankcard
USGS
Remote
QTime
SN
IPDS
GVP
CGE
DOI Talent
swam
All Bookmarks
jupyter
Status: Connected
Remote Clipboard
Hub Control Panel
Applications
Tue 11 Jun, 21:16
Default user
File System
Home
QGIS
Paraview
To get back to your files and main VICTOR control panel, open the previous browser tab


jupyter
Status: Connected
Remote Clipboard
Hub Control Panel
Applications
Wed 5 Jun, 16:20 Default user
File System
Home
QGIS
Paraview
Click on QGIS, if you get an error message, choose “launch anyway”

Applications | LaharZ 2.1.1 | Untitled Project — QGIS | Terminal | Tue 11 Jun, 23:04 Default use

Untitled Project — QGIS

Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help

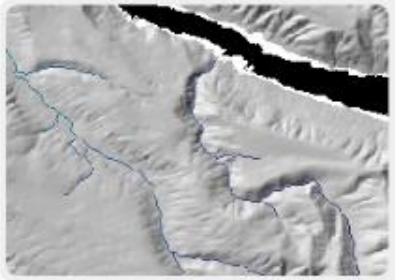
Click on “New Project”

Browser

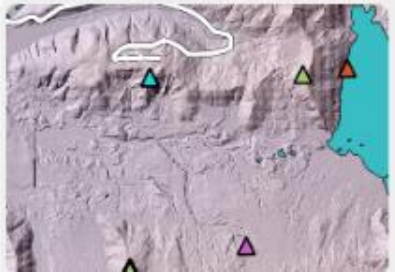
Favorites
Spatial Bookmarks
Home
/
GeoPackage
SpatiaLite
PostgreSQL
MS SQL Server
WMS/WMTS
Vector Tiles
XYZ Tiles
WCS
WFS / OGC API - Features
ArcGIS REST Servers



ol
yan/Laharz/laharz/Licancabur/large_vol.qgz
EPSG:4326 (EPSG:4326 - WGS 84)

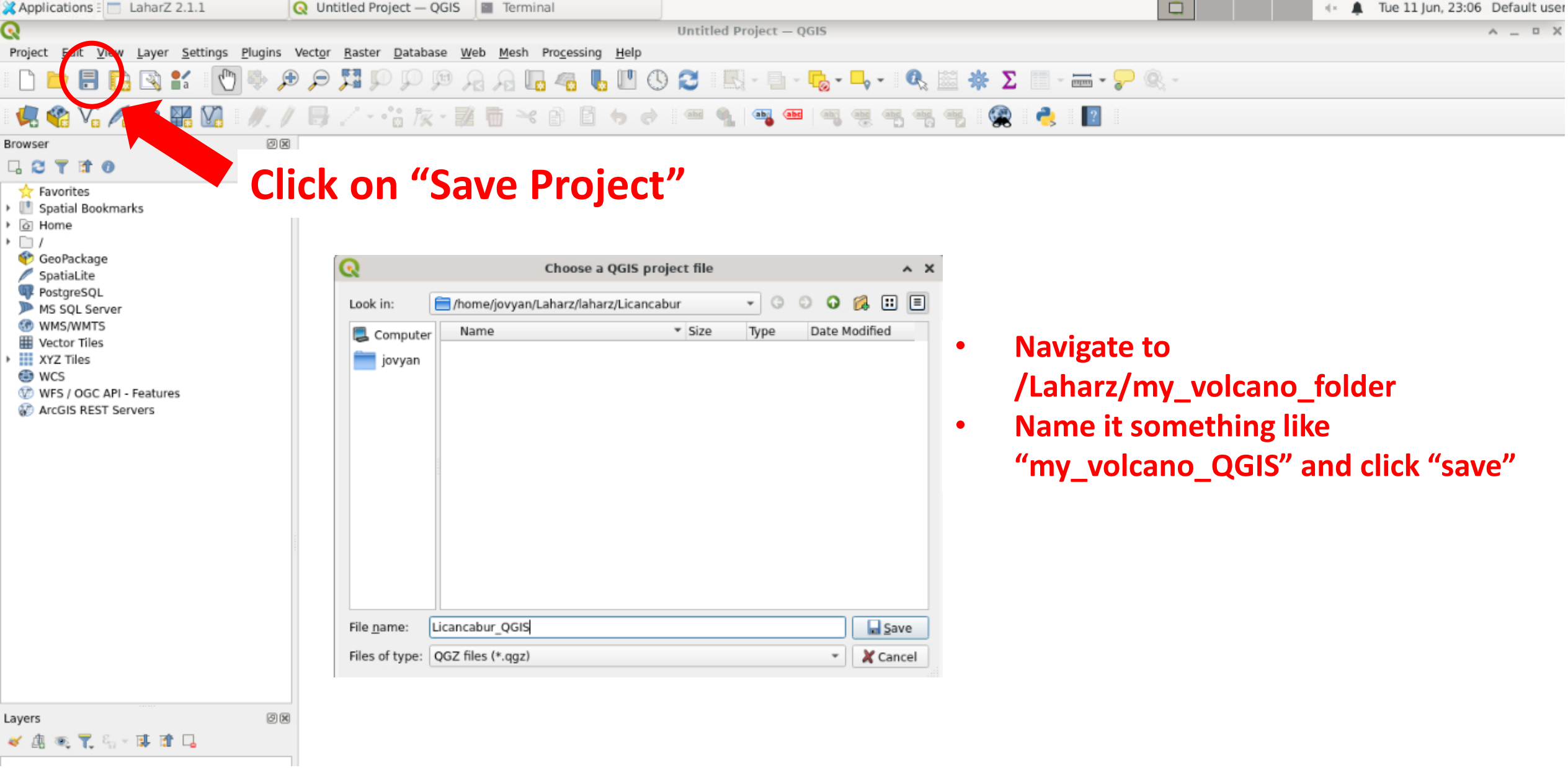


wgs_outputs
/home/jovyan/Laharz/laharz/rainier/wgs_outputs.qgz
EPSG:4326 (EPSG:4326 - WGS 84)

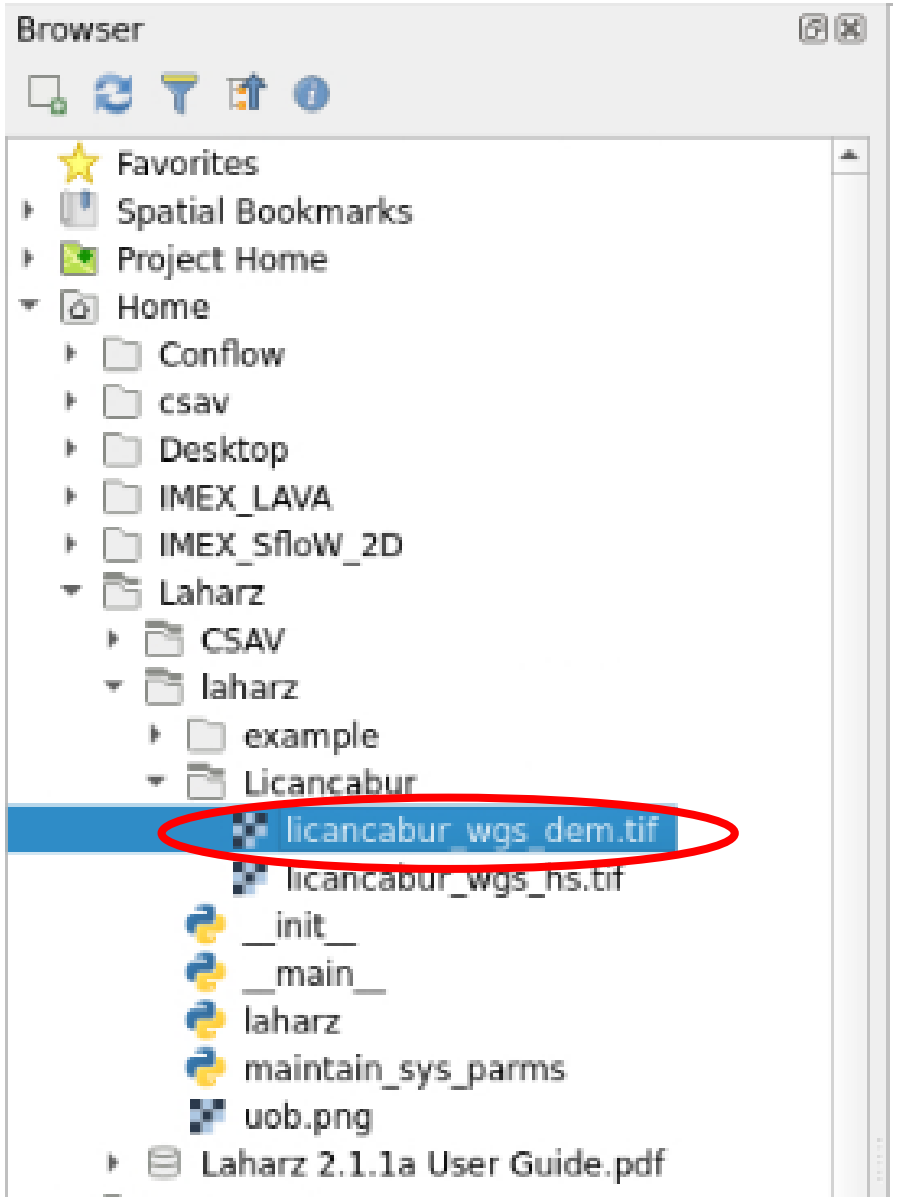


MSH_GIS
/home/jovyan/Laharz/GIS_exercise/MSH_GIS.qgz
EPSG:26710 (EPSG:26710 - NAD27 / UTM zone 10N)

Project Templates

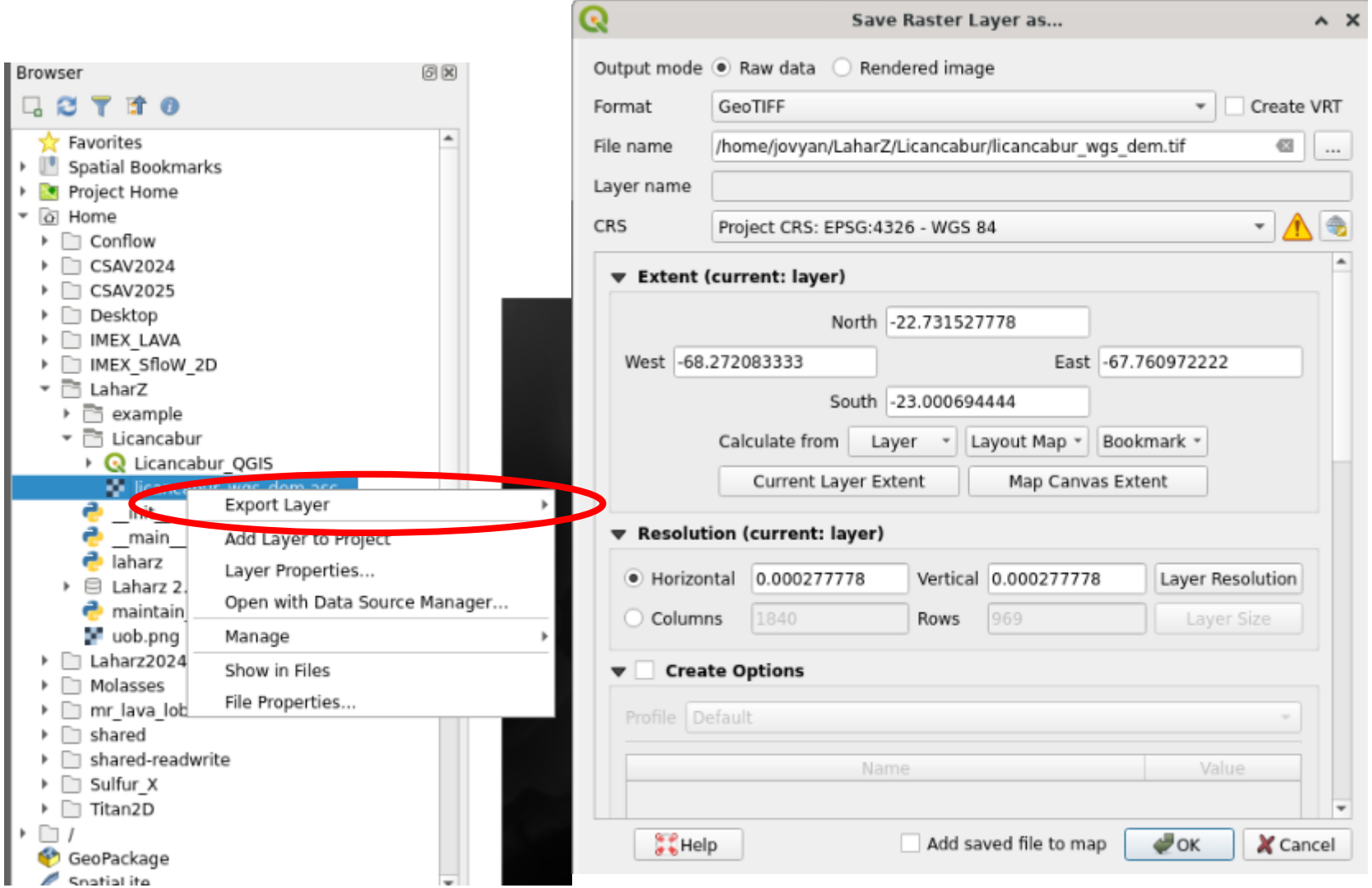


- Navigate to /Laharz/my_volcano_folder
- Name it something like “my_volcano_QGIS” and click “save”

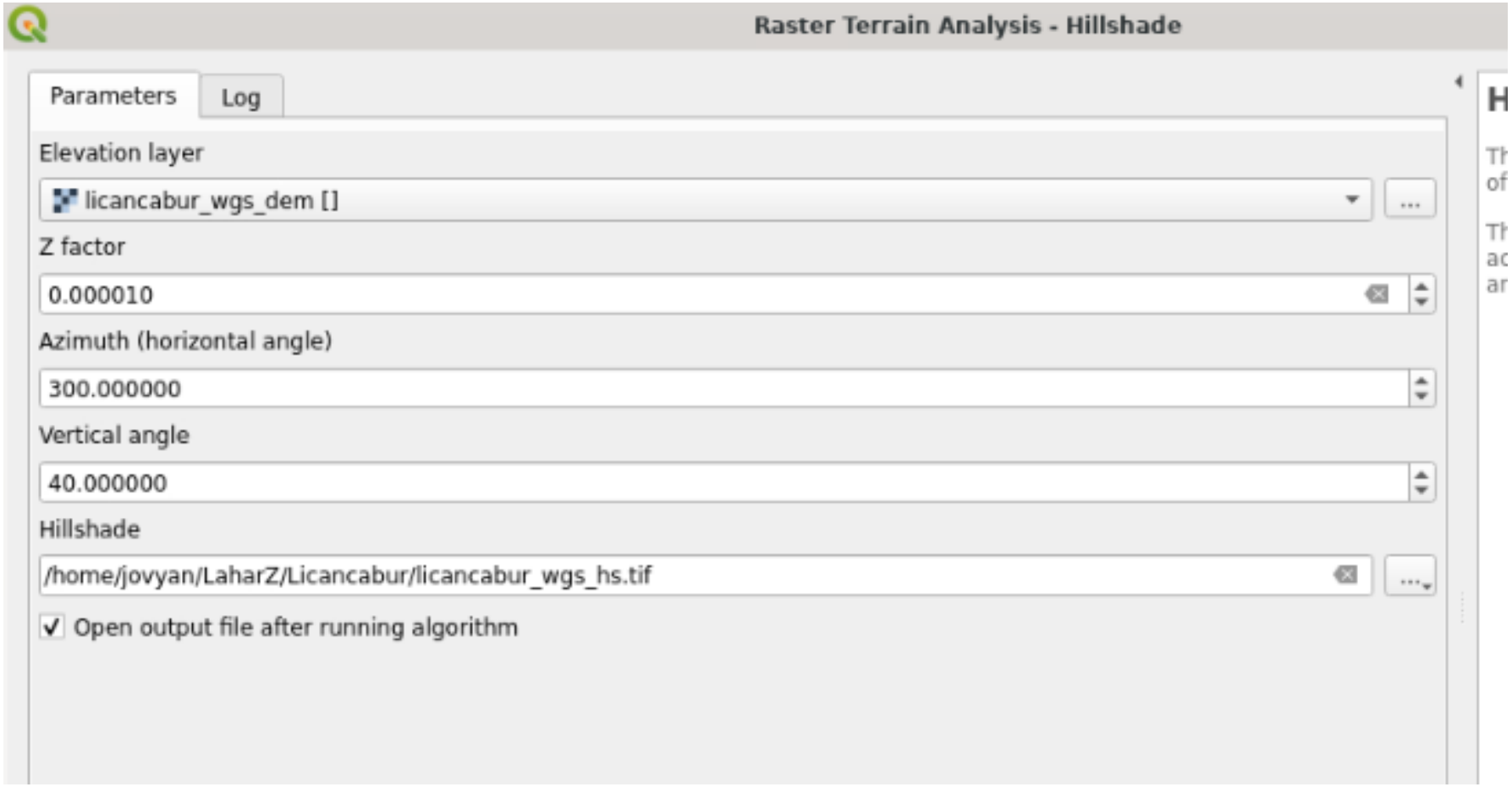


- **Navigate to the Laharz/my_volcano_folder**
- **Drag and drop your DEM file onto your map**

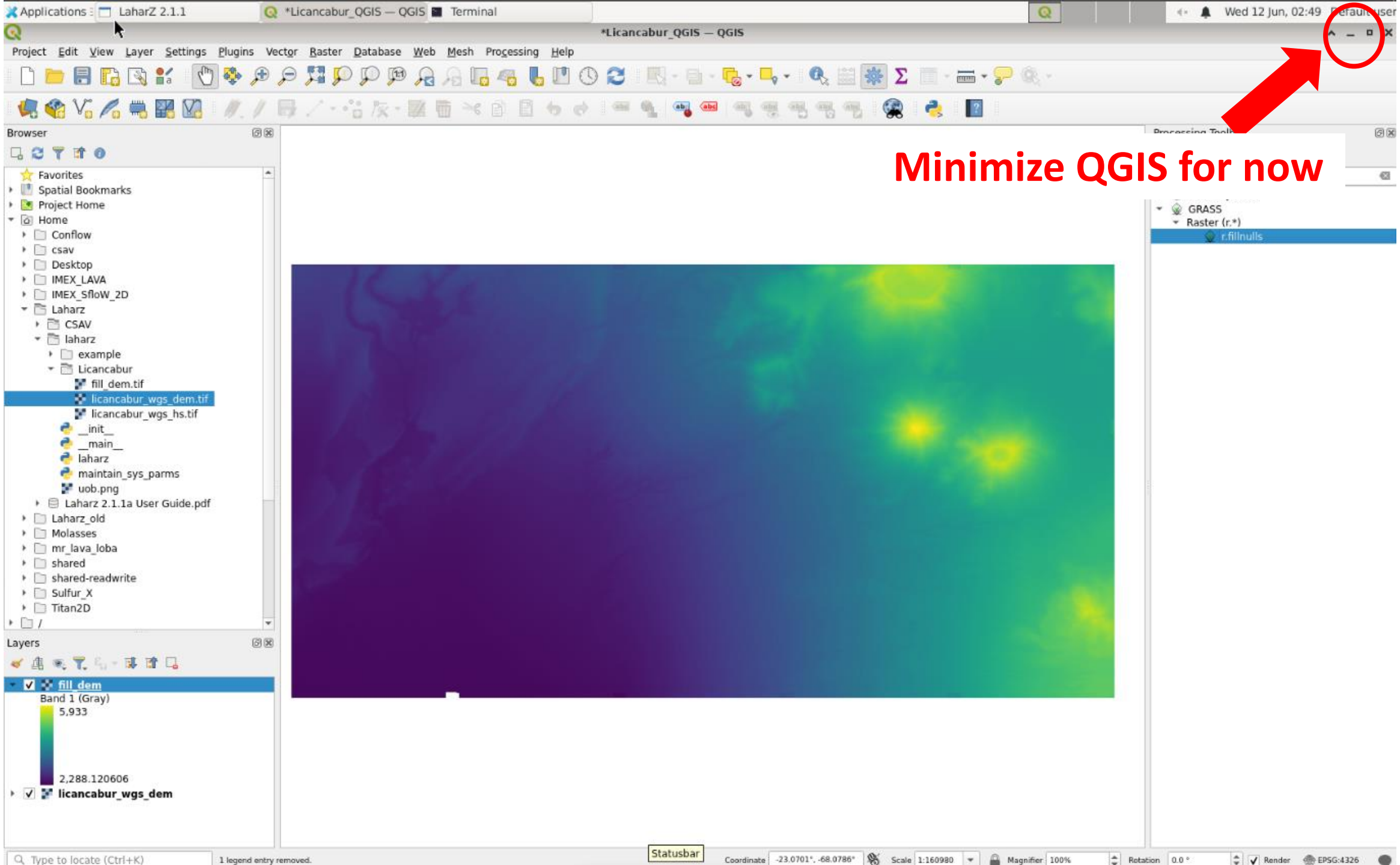
(the image shows the Licancabur example)

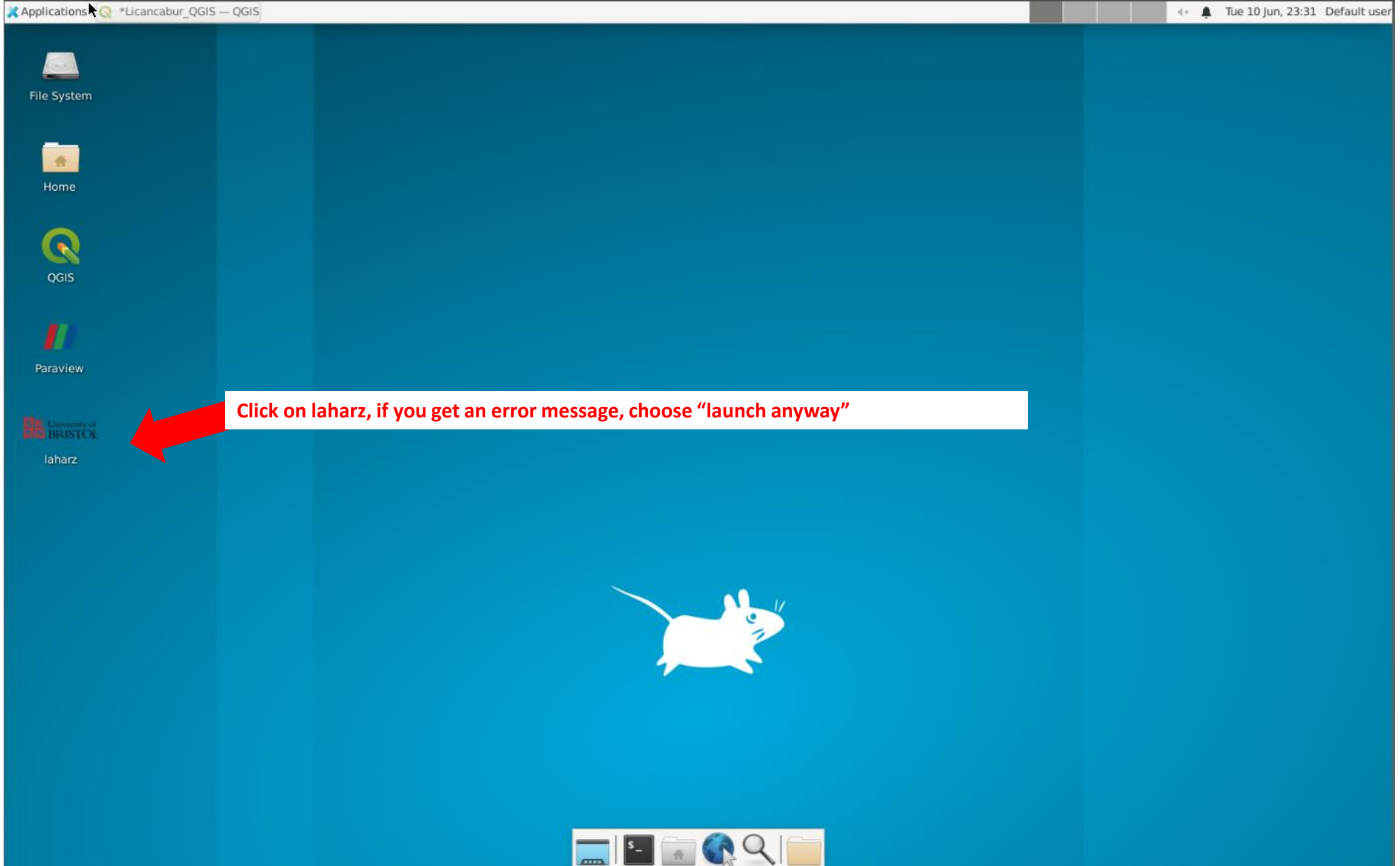


- Navigate to the DEM in the QGIS browser
- Right click on the DEM and choose “Export Layer”
- We will export the DEM as a GeoTiff.
- Save the file in /LaharZ/my_volcano_folder as “my_volcano_dem.tif”

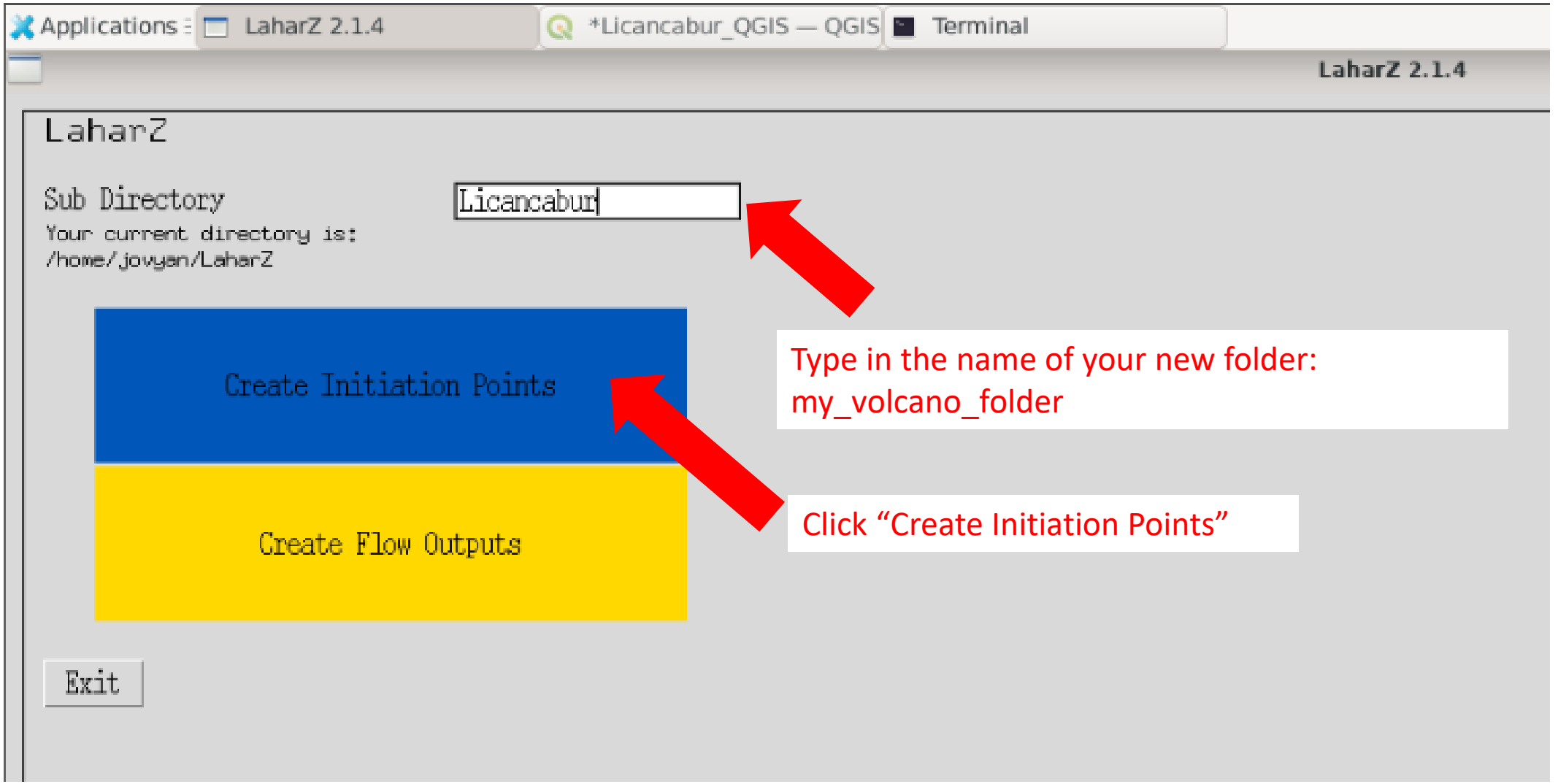


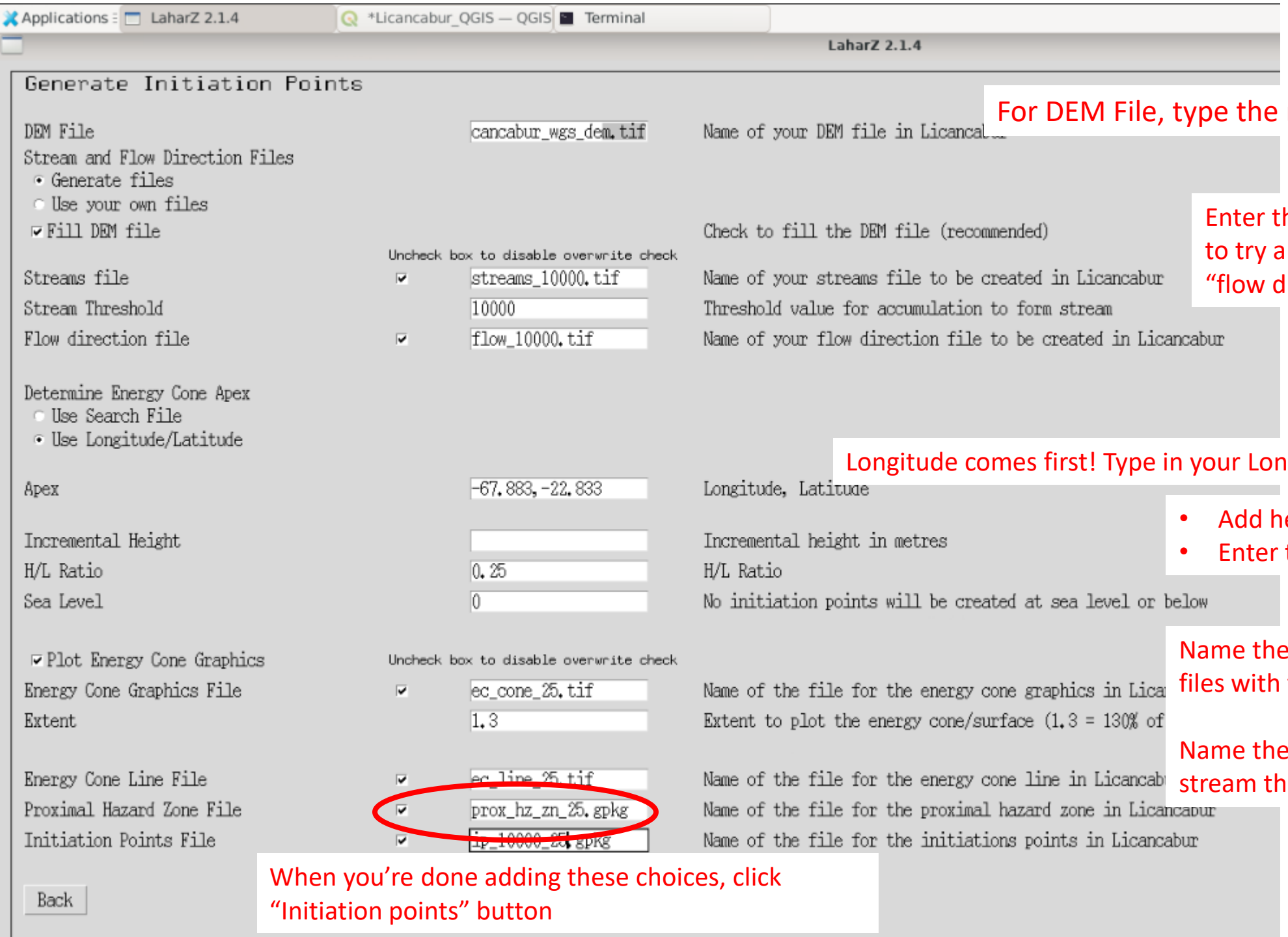
- **Now make a hillshade of the DEM**
- **Open the Raster Terrain Analysis →Hillshade tool**
- **Remember to change the Z factor to 0.00001 because the raster is in degrees lat/long**
- **Save the file in /LaharZ/my_volcano_folder as “my_volcano_hs.tif”**


Applications
LaharZ 2.1.1
*Licancabur_QGIS — QGIS
Terminal
Wed 12 Jun, 02:49
*Licancabur_QGIS — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Minimize QGIS for now
Browser
Favorites
Spatial Bookmarks
Project Home
Home
Conflow
csav
Desktop
IMEX_LAVA
IMEX_SfloW_2D
Laharz
CSAV
laharz
example
Licancabur
fill_dem.tif
licancabur_wgs_dem.tif
licancabur_wgs_hs.tif
__init__
__main__
laharz
maintain_sys_parms
uob.png
Laharz 2.1.1a User Guide.pdf
Laharz_old
Molasses
mr_lava_loba
shared
shared-readwrite
Sulfur_X
Titan2D
/
Layers
fill_dem
Band 1 (Gray)
5,933
2,288.120606
licancabur_wgs_dem
GRASS
Raster (r.*)
r.fillnulls
Type to locate (Ctrl+K)
1 legend entry removed.
Statusbar
Coordinate -23.0701°, -68.0786°
Scale 1:160980
Magnifier 100%
Rotation 0.0 °
Render
EPSG:4326


Applications
*Licancabur_QGIS — QGIS
Tue 10 Jun, 23:31
Default user
File System
Home
QGIS
Paraview
laharz
Click on laharz, if you get an error message, choose “launch anyway”

Let’s make a stream network, an H/L cone, and start points





For DEM File, type the name of your DEM (the tif)

Enter the stream threshold you decided to try and name the “streams file” and “flow direction” file using that number

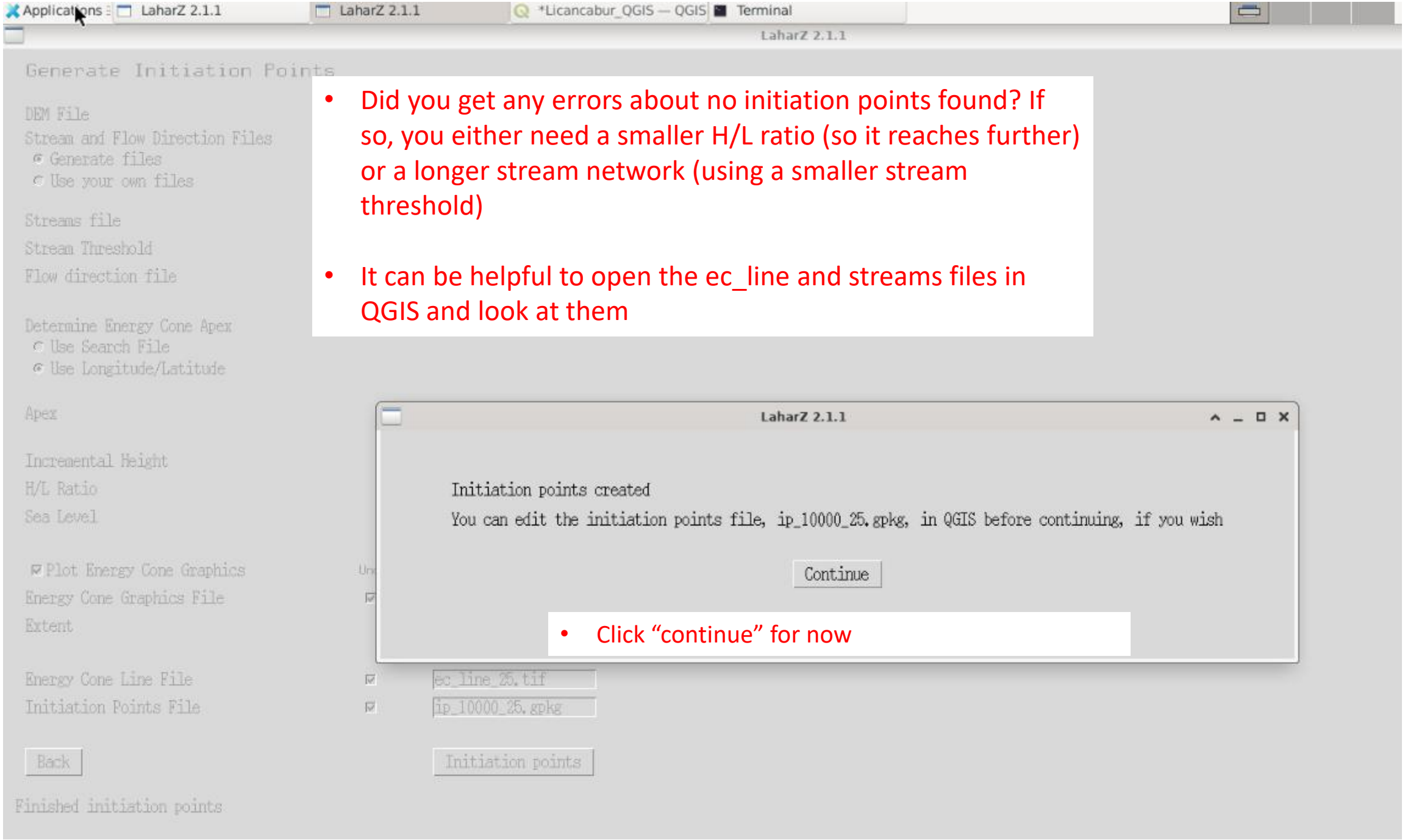
Longitude comes first! Type in your Longitude, Latitude

- Add height if you need to
- Enter the H/L ratio you decided to try

Name the energy cone and energy line files with the H/L ratio used

Name the initiation points with the stream threshold and H/L ratio you used

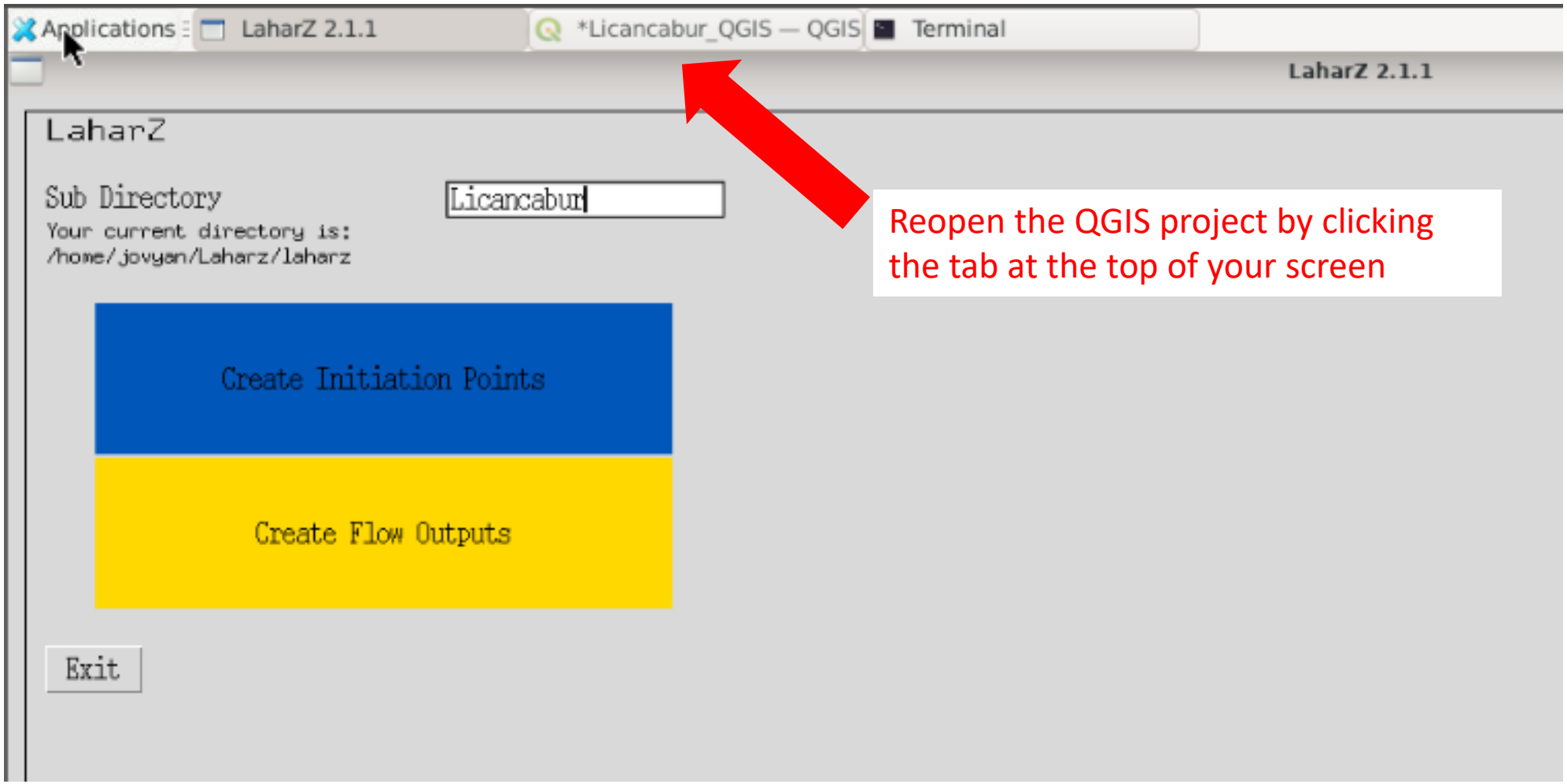
When you’re done adding these choices, click “Initiation points” button

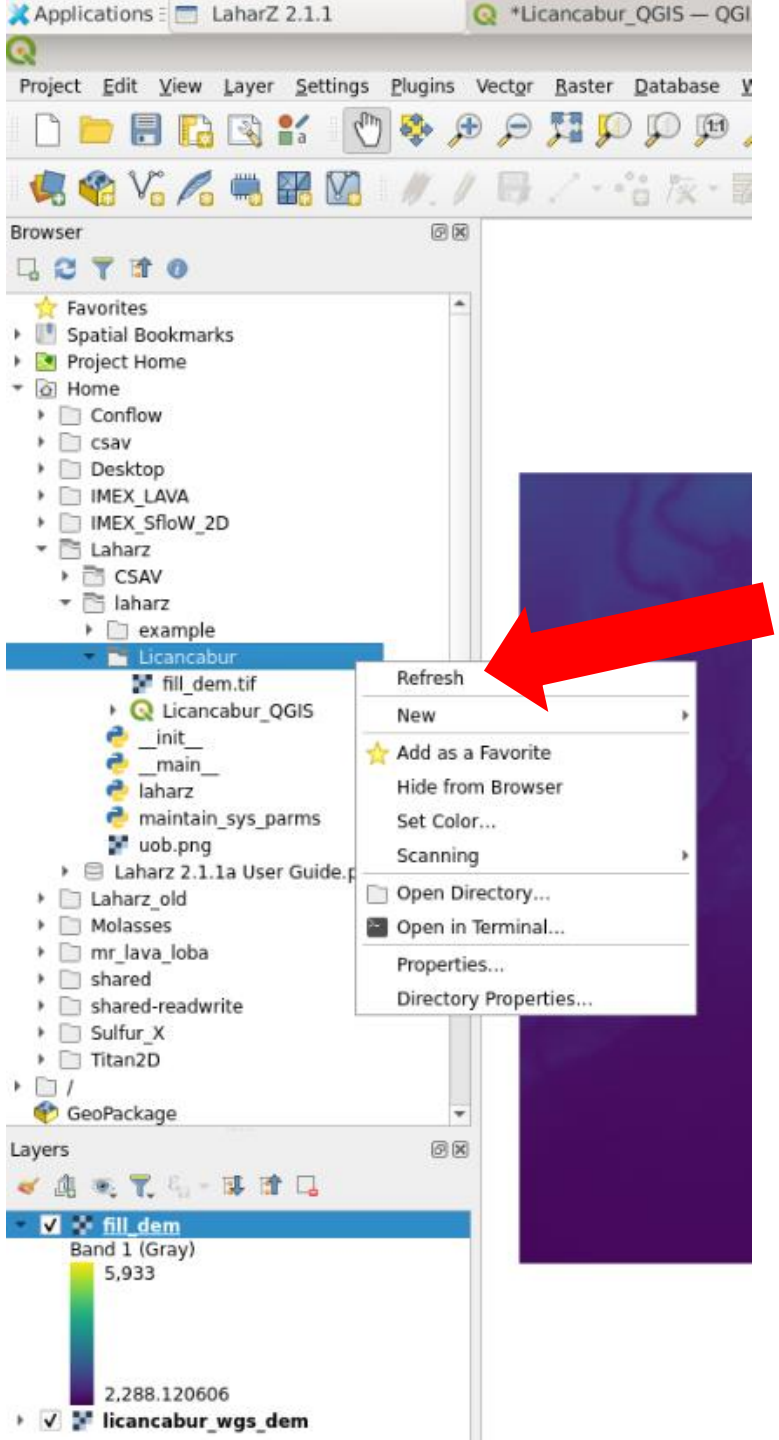


- Did you get any errors about no initiation points found? If so, you either need a smaller H/L ratio (so it reaches further) or a longer stream network (using a smaller stream threshold)

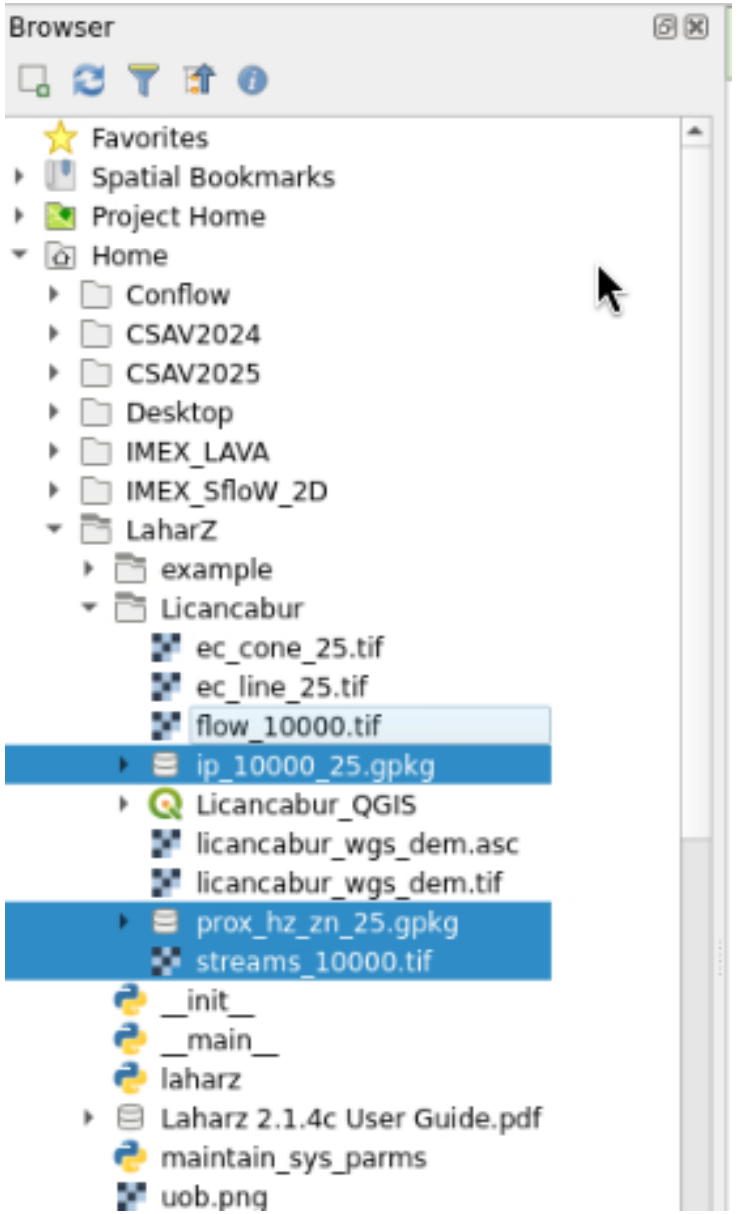
- It can be helpful to open the ec_line and streams files in QGIS and look at them

- Click “continue” for now


Applications
LaharZ 2.1.1
*Licancabur_QGIS — QGIS
Terminal
LaharZ 2.1.1
LaharZ
Sub Directory
Licancabur
Your current directory is:
/home/jovyan/Laharz/laharz
Create Initiation Points
Create Flow Outputs
Exit
Reopen the QGIS project by clicking the tab at the top of your screen

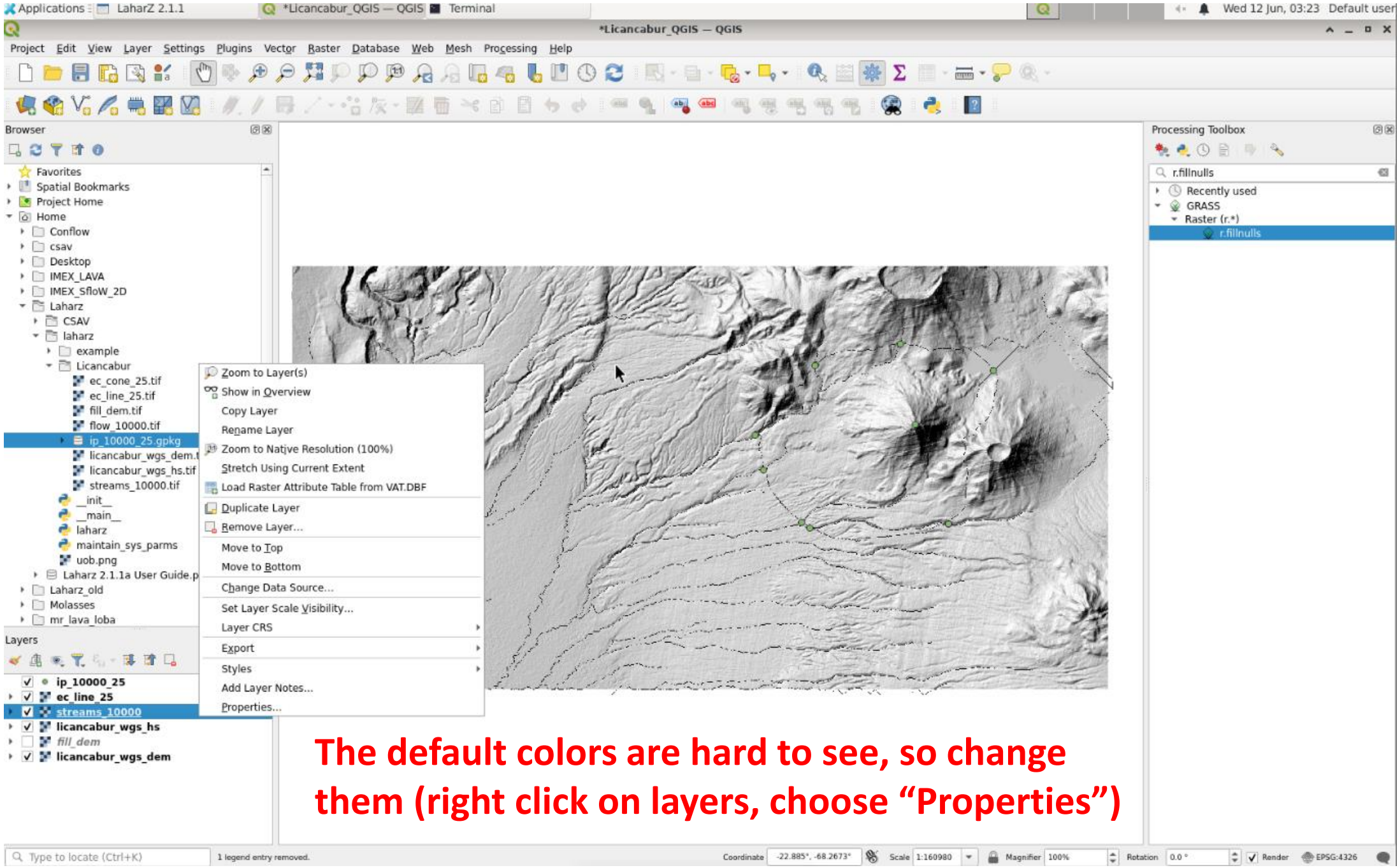


Right click on your volcano folder and click “Refresh” to see your new files



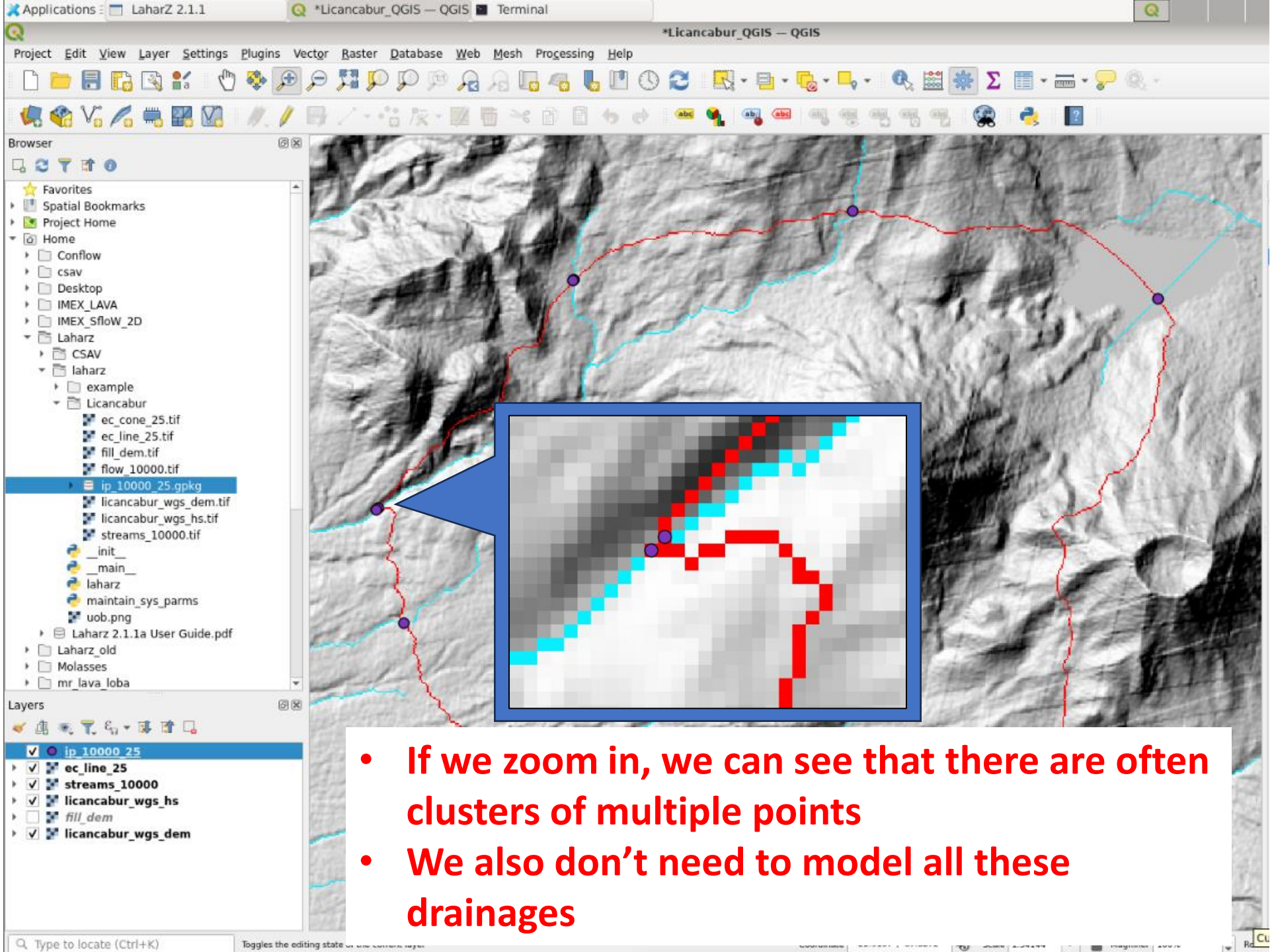
Let's look at the stream network, H/L cone (ec_line), and the initiation points we created!

Add these files to your map (for your new volcano):
streams_10000.tif
ec_line_25.tif
ip_10000_25.gpkg
prox_hz_zn_25.gpkg

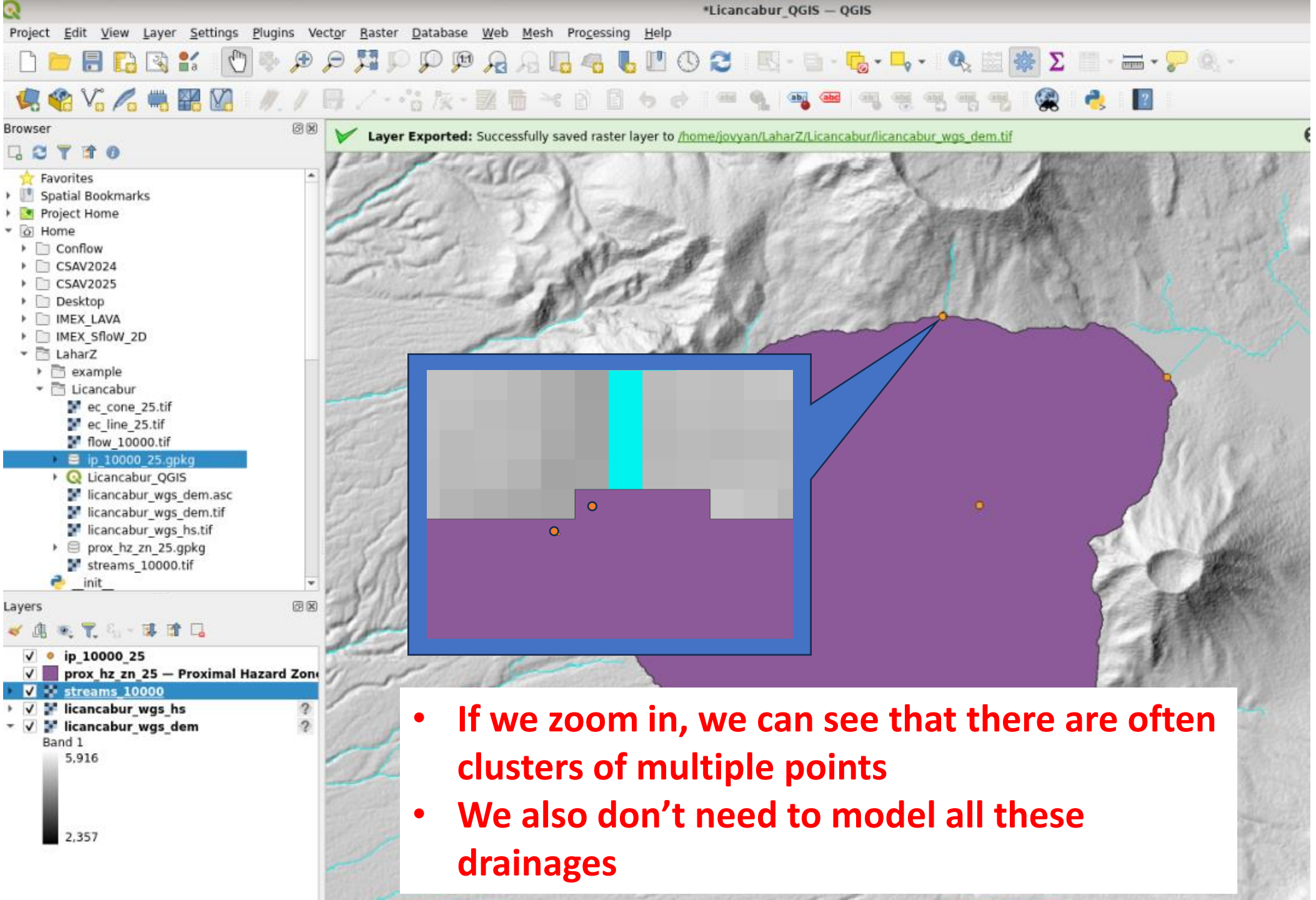

Applications
LaharZ 2.1.1
*Licancabur_QGIS — QGIS
Terminal
Wed 12 Jun, 03:23
Default user
*Licancabur_QGIS — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Browser
Favorites
Spatial Bookmarks
Project Home
Home
Conflow
csav
Desktop
IMEX_LAVA
IMEX_SfloW_2D
Laharz
CSAV
laharz
example
Licancabur
ec_cone_25.tif
ec_line_25.tif
fill_dem.tif
flow_10000.tif
ip_10000_25.gpkg
licancabur_wgs_dem.t
licancabur_wgs_hs.tif
streams_10000.tif
__init__
__main__
laharz
maintain_sys_parms
uob.png
Laharz 2.1.1a User Guide.p
Laharz_old
Molasses
mr_lava_loba
Zoom to Layer(s)
Show in Overview
Copy Layer
Rename Layer
Zoom to Native Resolution (100%)
Stretch Using Current Extent
Load Raster Attribute Table from VAT.DBF
Duplicate Layer
Remove Layer...
Move to Top
Move to Bottom
Change Data Source...
Set Layer Scale Visibility...
Layer CRS
Export
Styles
Add Layer Notes...
Properties...
Layers
ip_10000_25
ec_line_25
streams_10000
licancabur_wgs_hs
fill_dem
licancabur_wgs_dem
Processing Toolbox
r.fillnulls
Recently used
GRASS
Raster (r.*)
r.fillnulls
The default colors are hard to see, so change them (right click on layers, choose “Properties”)
Type to locate (Ctrl+K)
1 legend entry removed.
Coordinate -22.885°, -68.2673°
Scale 1:160980
Magnifier 100%
Rotation 0.0 °
Render
EPSG:4326

Adjust if needed!

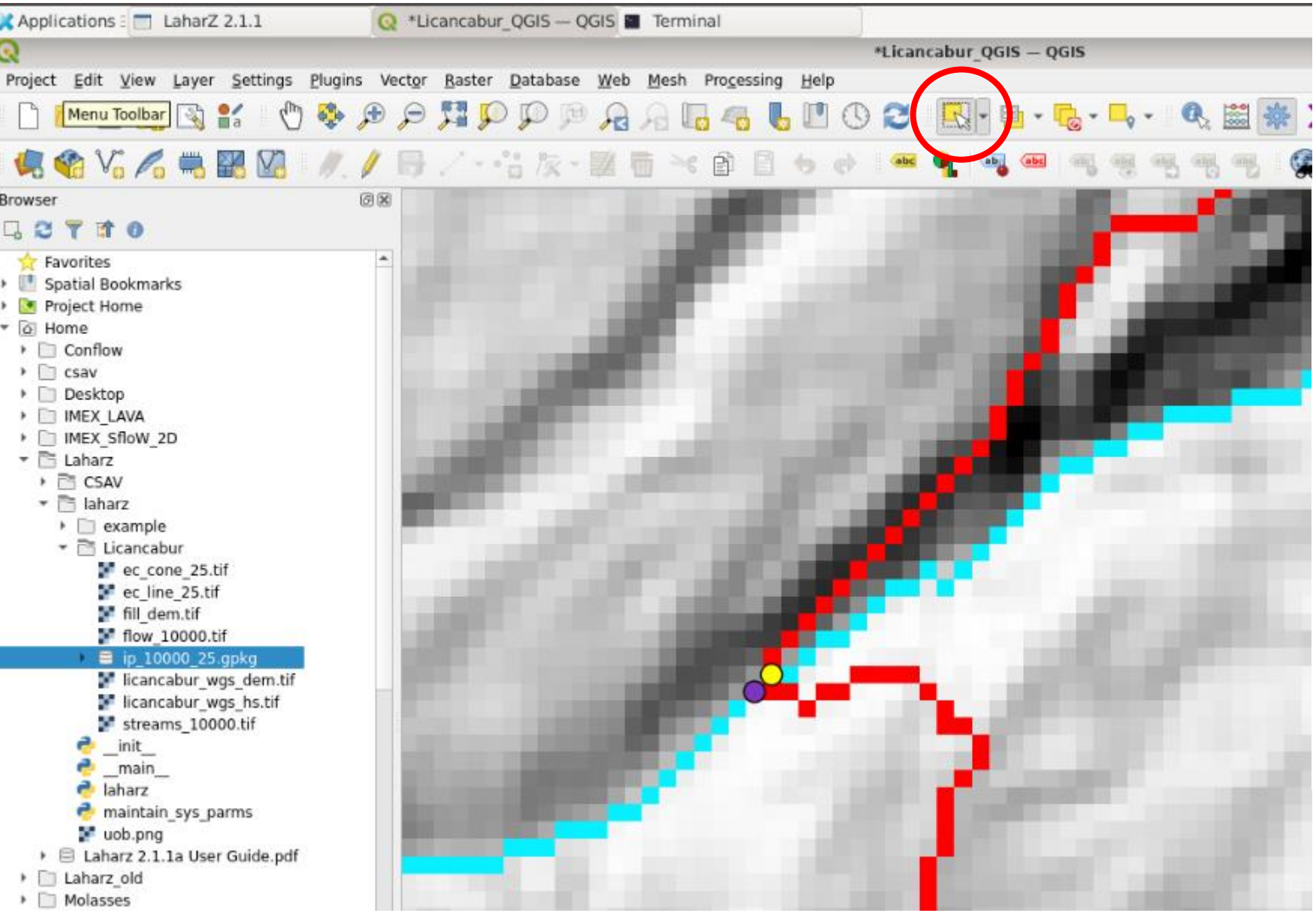
- Examine the H/L cone (ec_line), the stream network (streams), and the initiation points (ip)
- Did you get any errors about no initiation points found? If so, you either need a smaller H/L ratio (so it reaches further) or a longer stream network (using a smaller stream threshold)
- Compare the stream network to any topographical or geological maps you have
- Does the stream network look good?
  - Is it too short? If so, try a smaller stream threshold number
  - Is it too long with lots of crazy branches? Try a larger stream threshold number
- Are there too many initiation points? Are they clustered? Delete some!
- Ask me questions!



- If we zoom in, we can see that there are often clusters of multiple points
- We also don’t need to model all these drainages

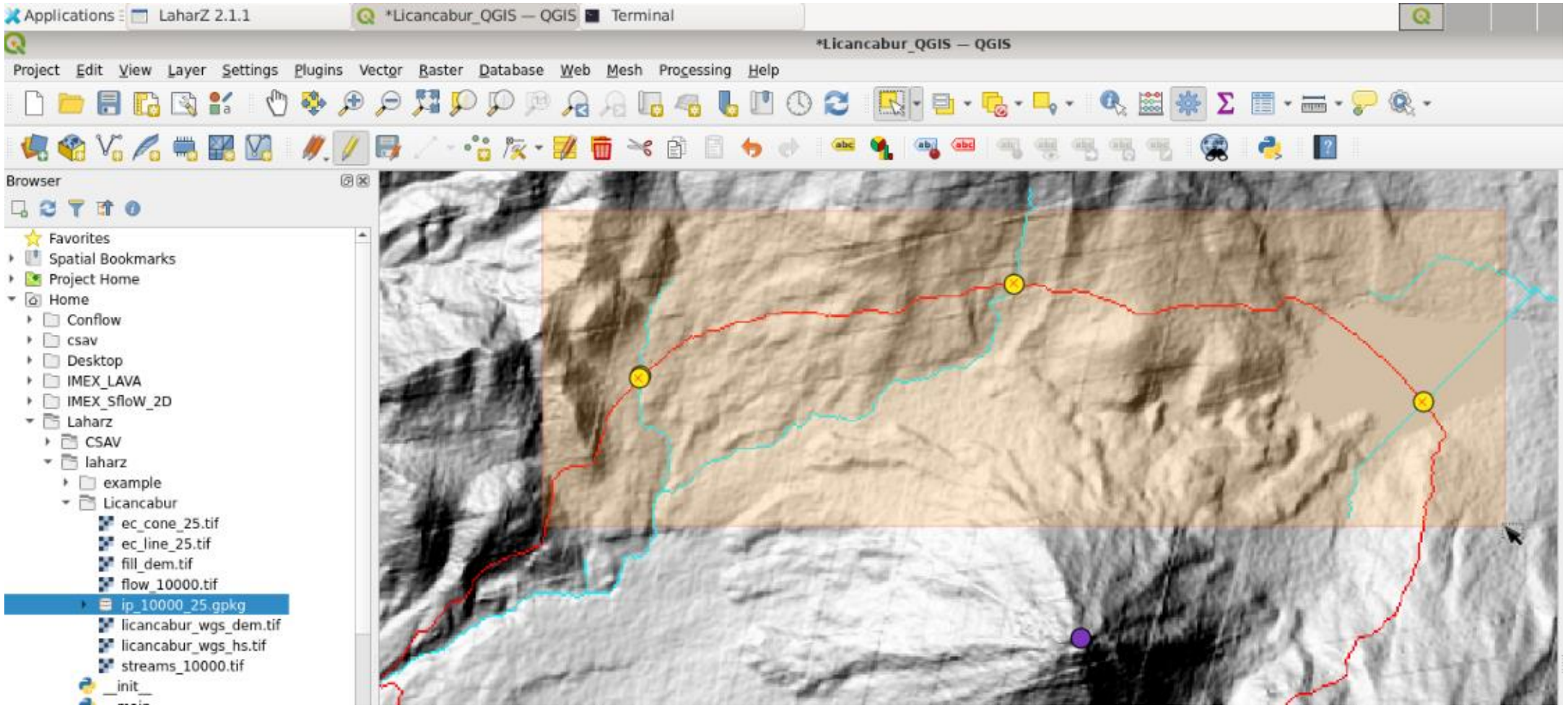


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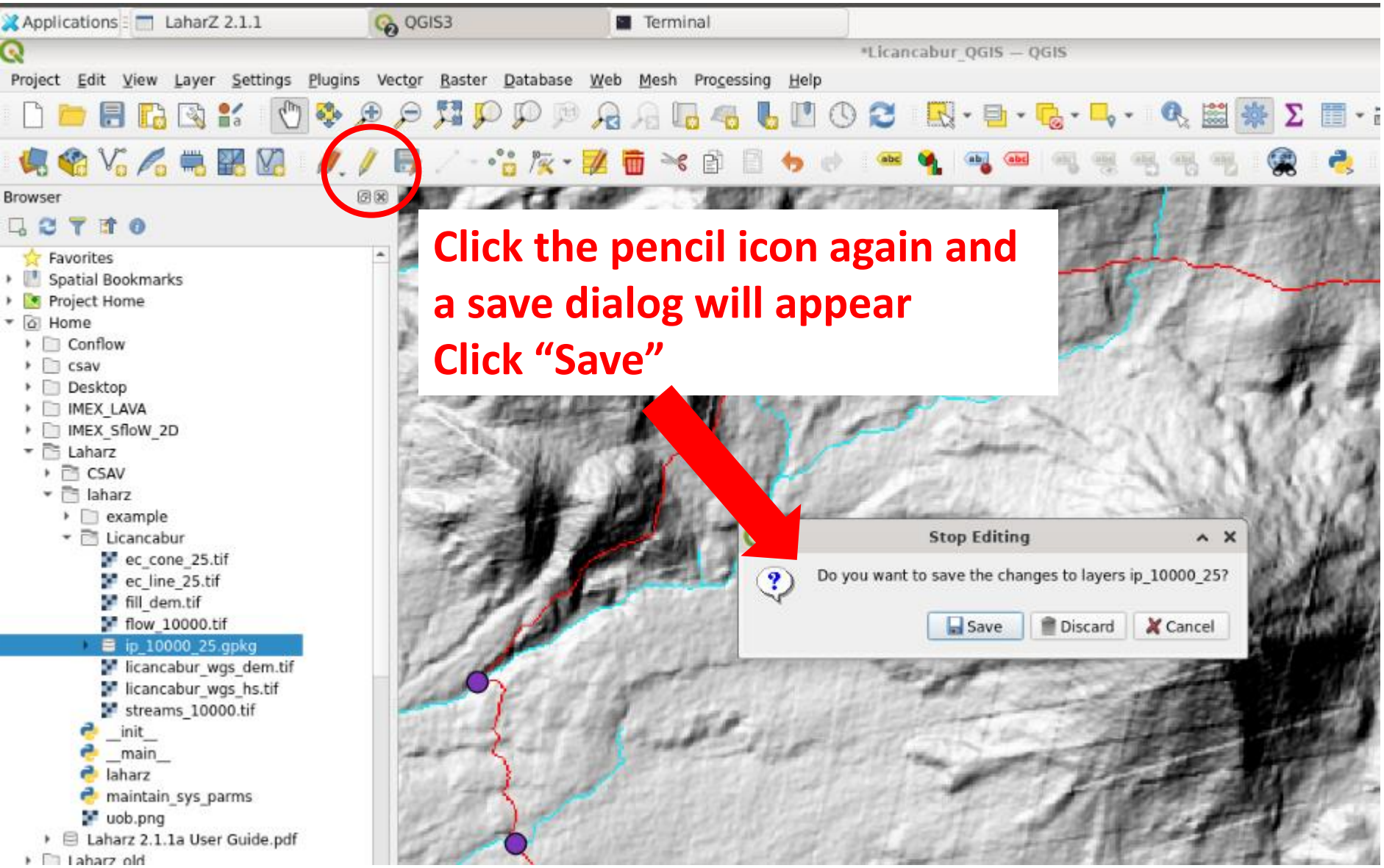
Let's delete some points

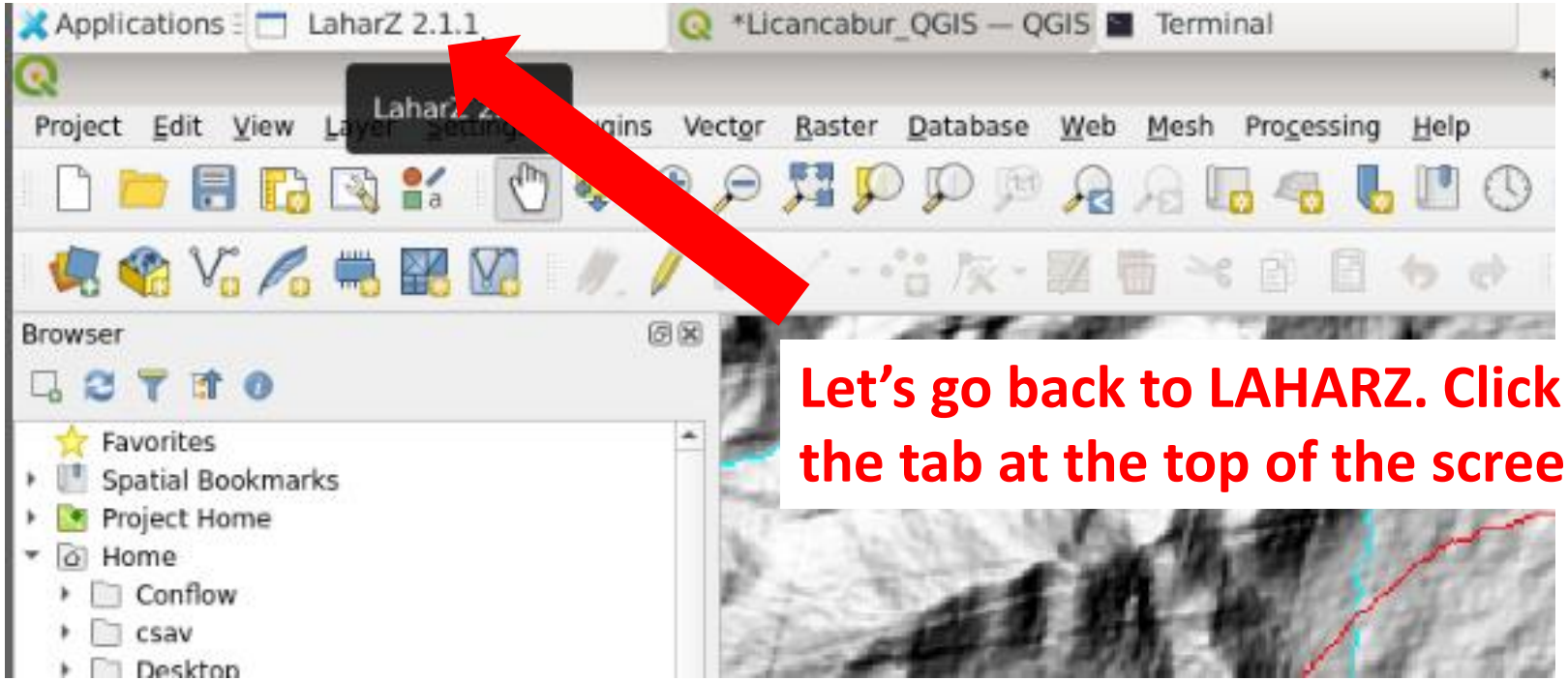
- Turn on the "Select features by area" tool and zoom in to each point cluster
- Click on the point you want to delete
- It will turn yellow
- The press the "delete" key on your keyboard



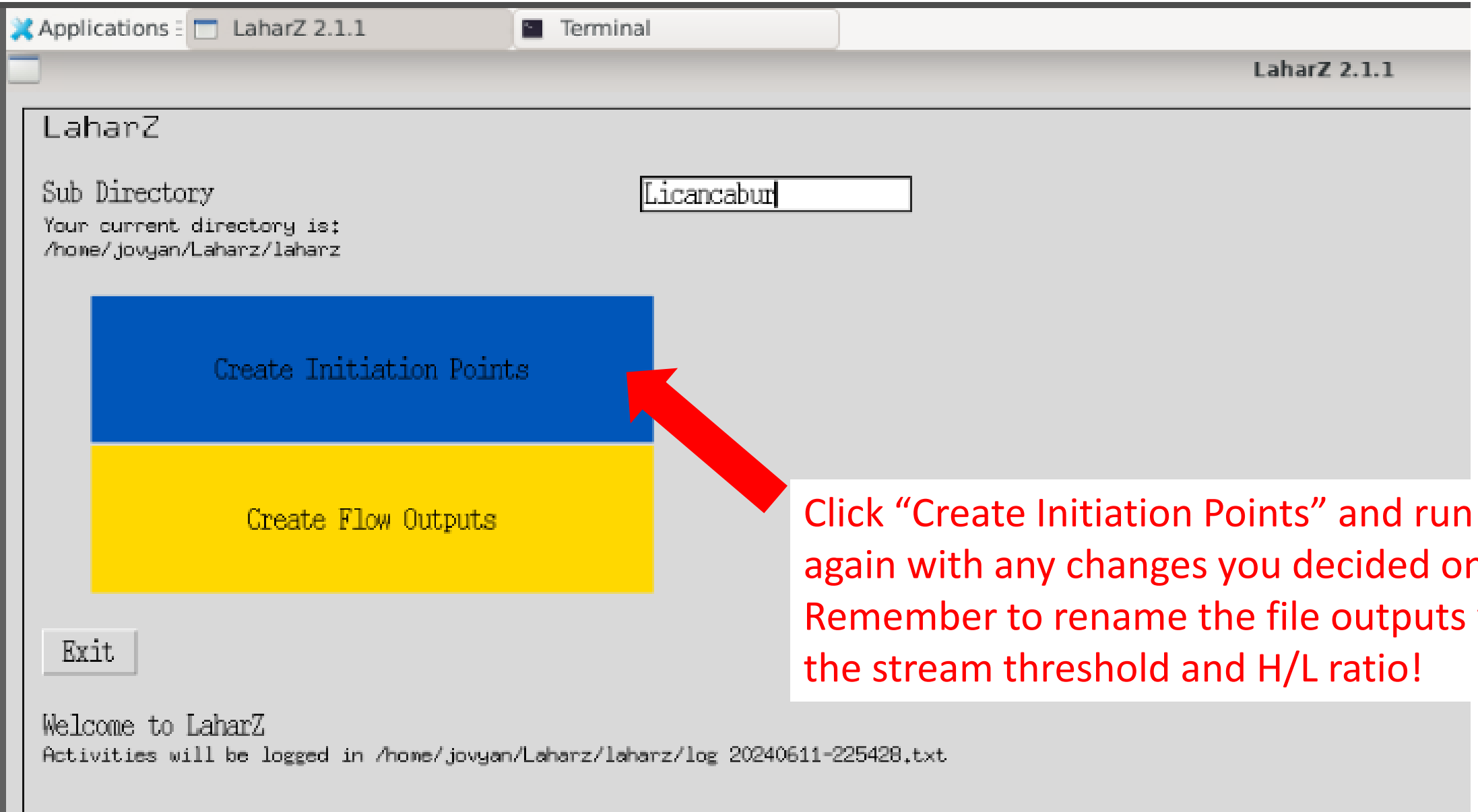
Let's delete some points

- You can also use the "Select features by area" tool to drag a selection box around multiple points
- They will turn yellow
- The press the "delete" key on your keyboard

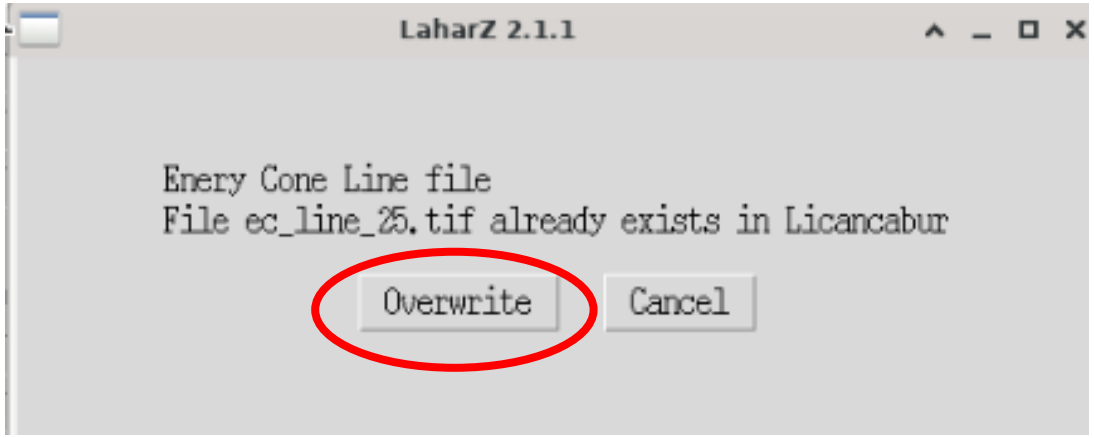

Applications
LaharZ 2.1.1
QGIS3
Terminal
*Licancabur_QGIS — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Browser
Favorites
Spatial Bookmarks
Project Home
Home
Conflow
csav
Desktop
IMEX_LAVA
IMEX_SfloW_2D
Laharz
CSAV
laharz
example
Licancabur
ec_cone_25.tif
ec_line_25.tif
fill_dem.tif
flow_10000.tif
ip_10000_25.gpkg
licancabur_wgs_dem.tif
licancabur_wgs_hs.tif
streams_10000.tif
__init__
__main__
laharz
maintain_sys_parms
uob.png
Laharz 2.1.1a User Guide.pdf
Laharz_old
Click the pencil icon again and
a save dialog will appear
Click “Save”
Stop Editing
Do you want to save the changes to layers ip_10000_25?
Save
Discard
Cancel


Applications
LaharZ 2.1.1
*Licancabur_QGIS — QGIS
Terminal
Project Edit View Layer Setting ...gins Vector Raster Database Web Mesh Processing Help
Browser
Favorites
Spatial Bookmarks
Project Home
Home
Conflow
csav
Desktop
Let's go back to LAHARZ. Click the tab at the top of the screen

You can explore what a different stream threshold does and how different ΔH/L values change the proximal zone.

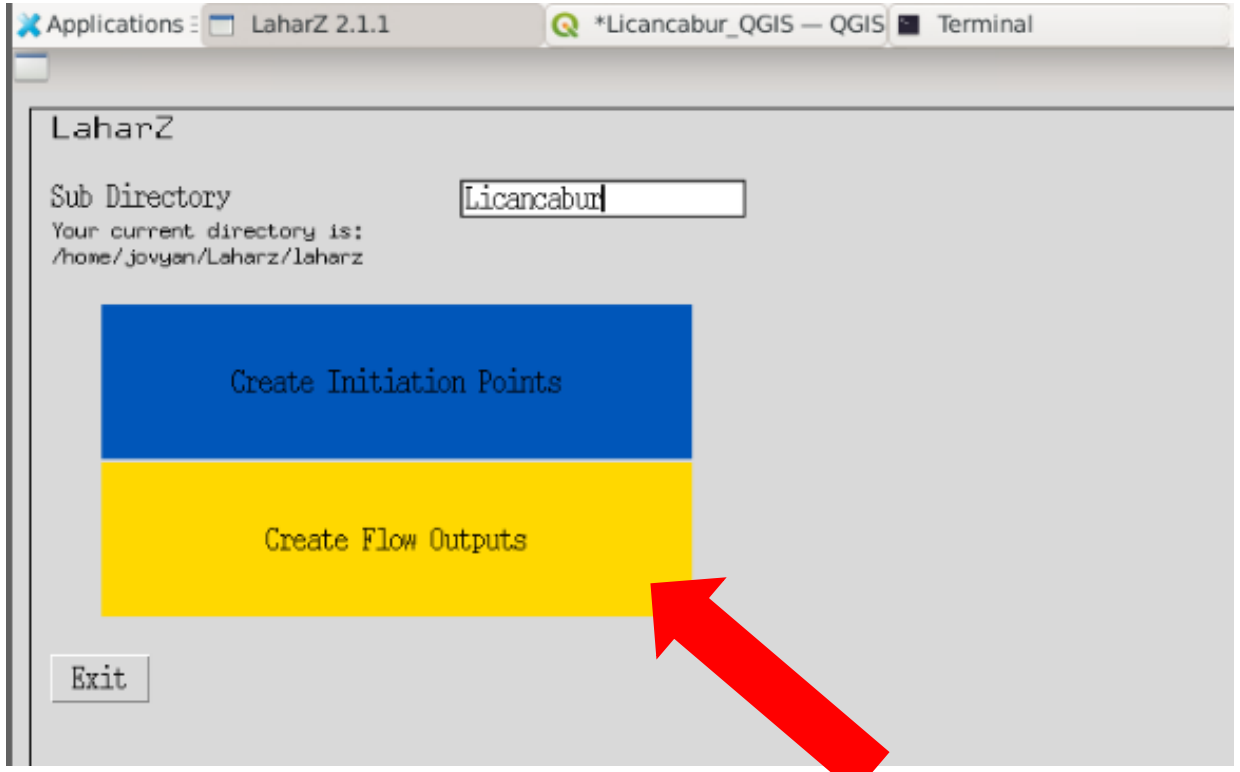


Click “Create Initiation Points” and run it again with any changes you decided on. Remember to rename the file outputs with the stream threshold and H/L ratio!

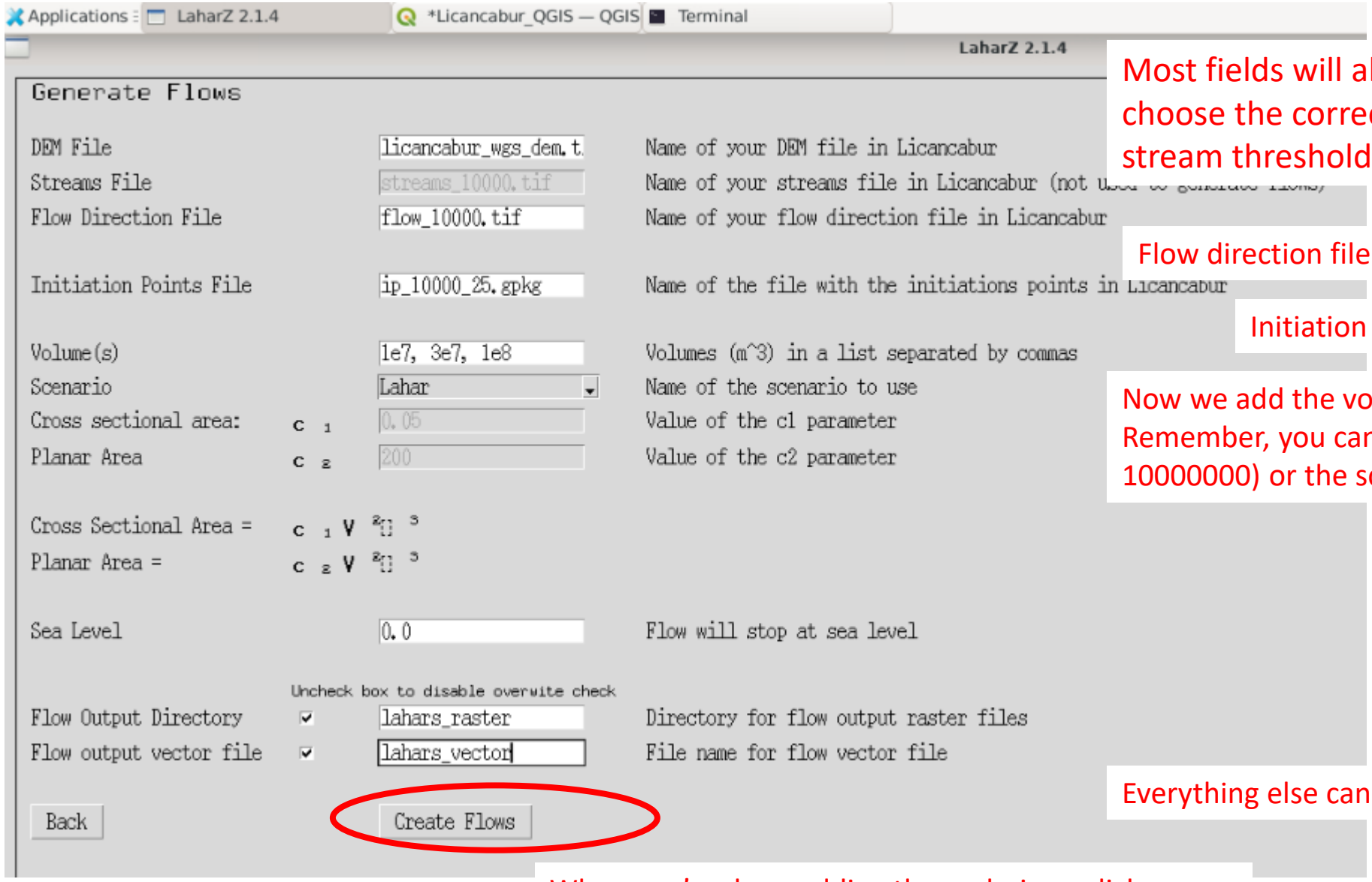


Choose “Overwrite” in the dialog boxes that pop up

Let’s run LAHARZ!



Click “Create Flow Outputs”



Most fields will already be filled in, but you need to choose the correct files that correspond to the stream threshold and H/L ratio you decided on (x)

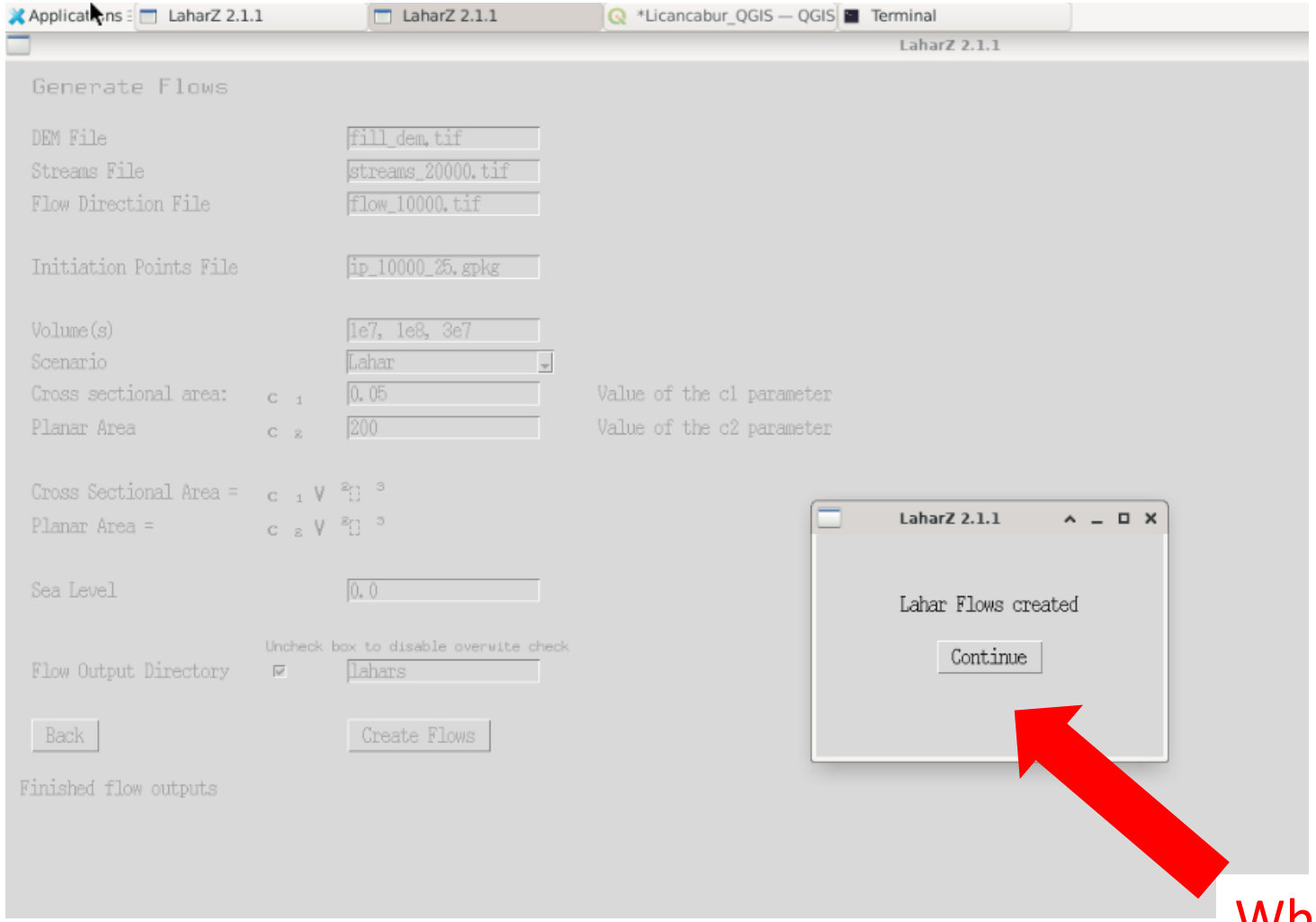
Flow direction file: flow_x.tif

Initiation points file: ip_x_x.gpkg

Now we add the volumes you decided on
Remember, you can type out the whole number (e.g., 10000000) or the scientific notation version (e.g., 1e7)

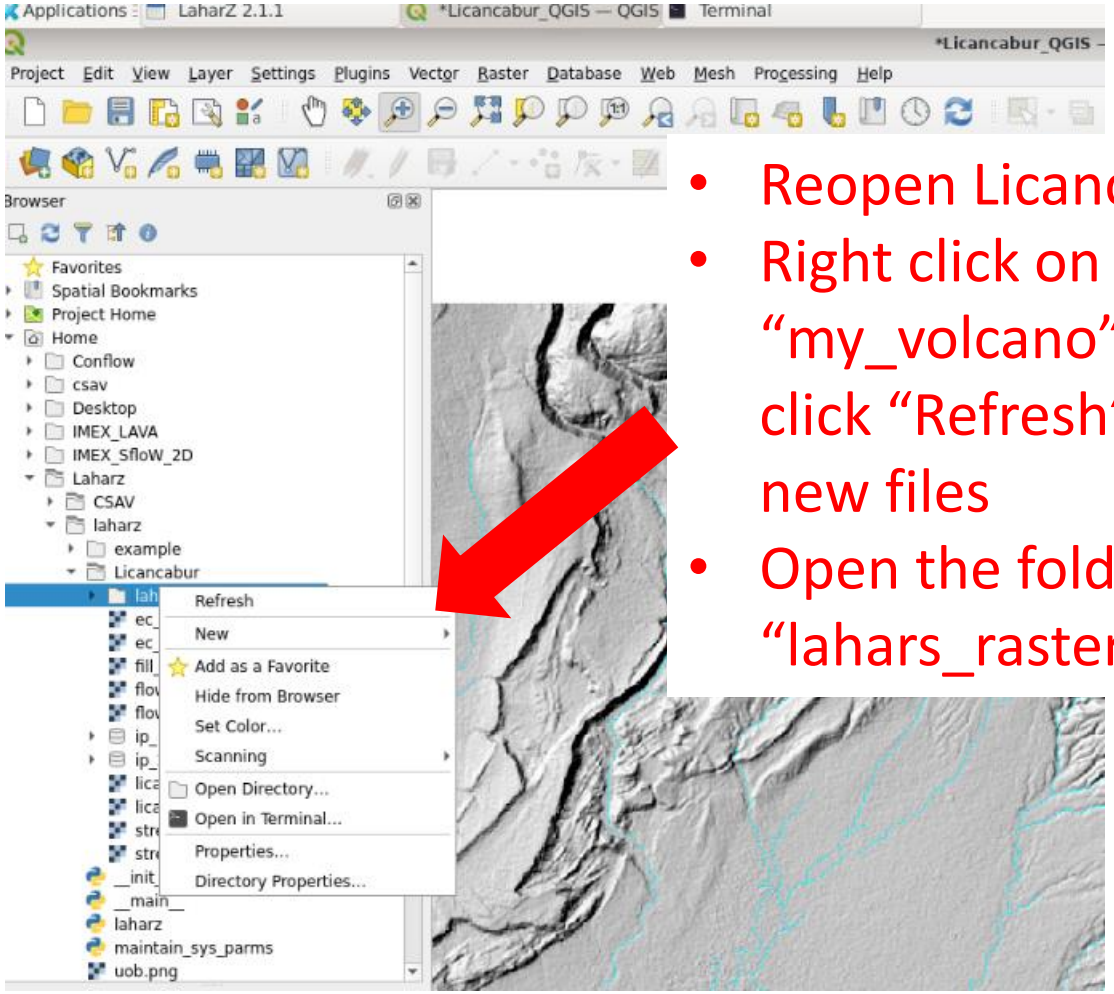
Everything else can stay as the default

When you're done adding these choices, click "Create flows" button

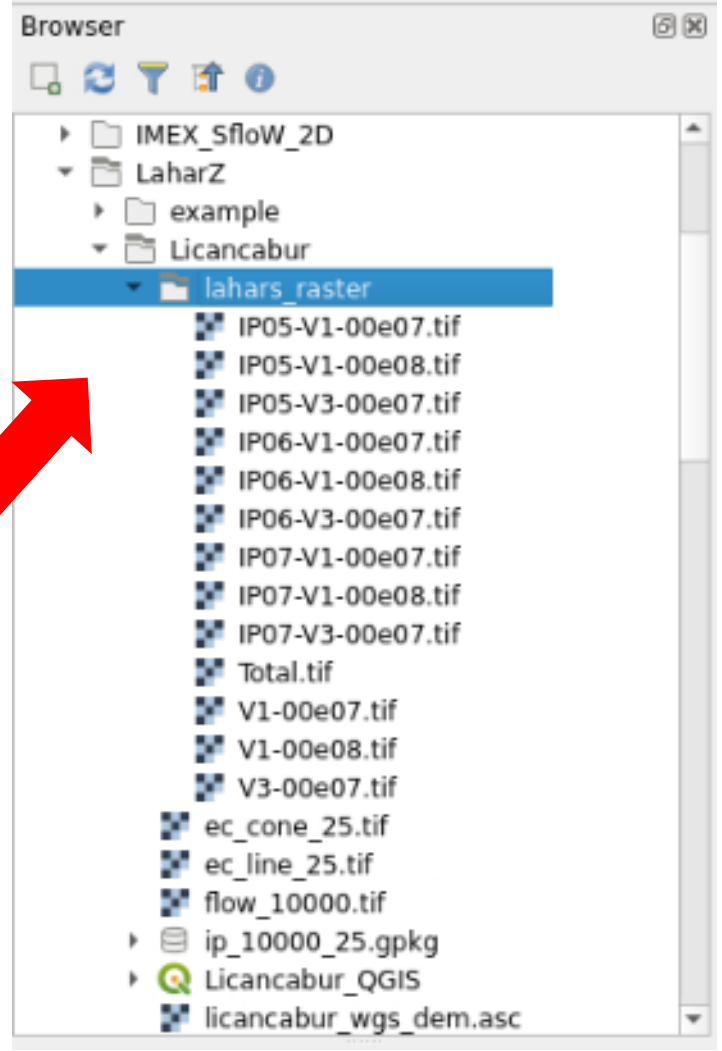


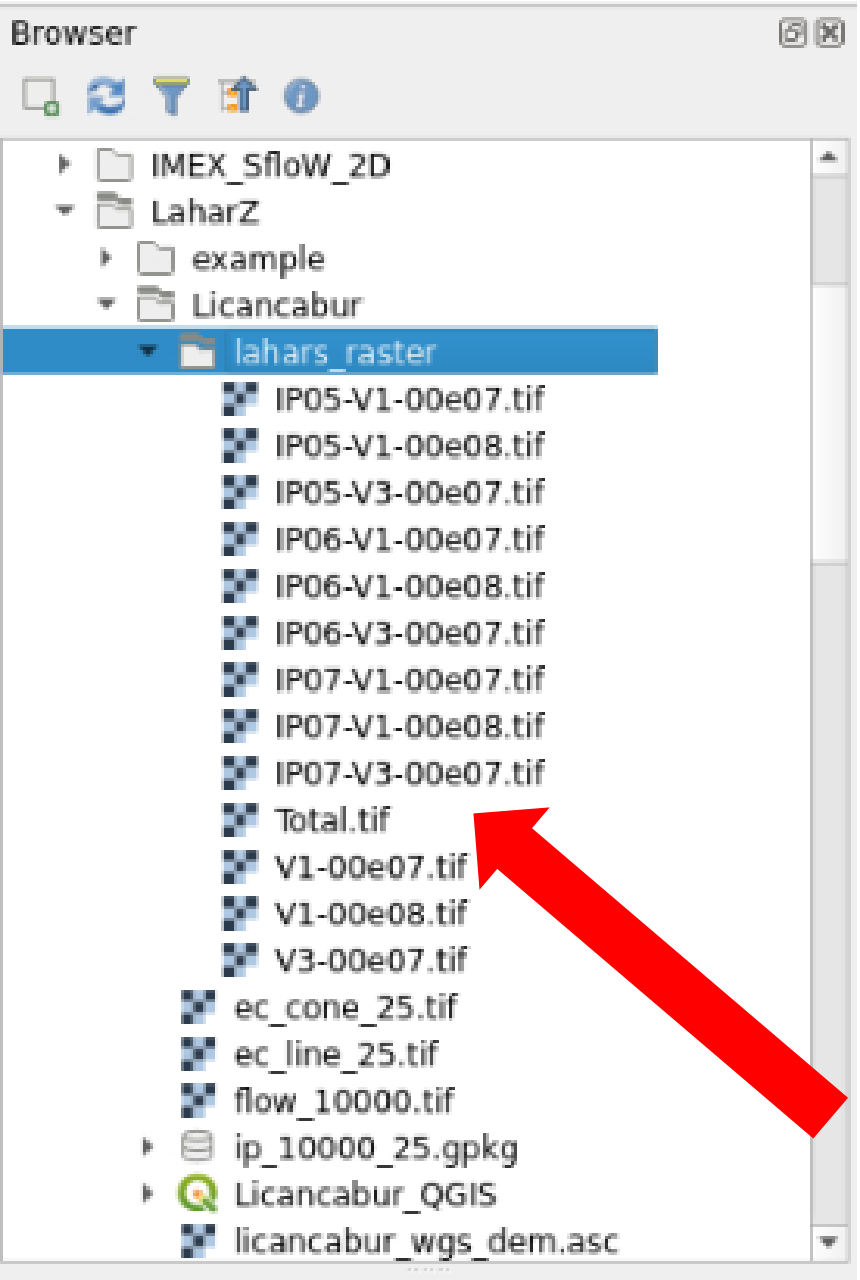
When it’s done, click “Continue”

Let’s look at the new files.

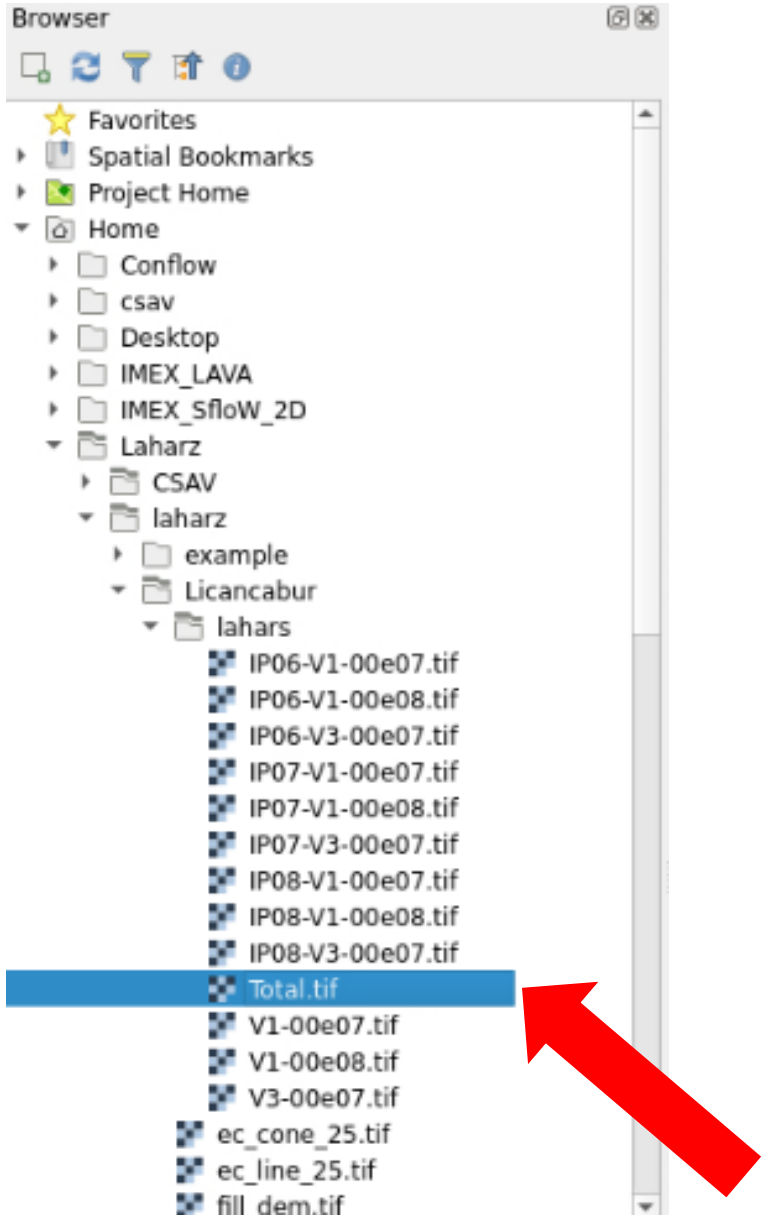


- Reopen Licancabur_QGIS
- Right click on the “my_volcano” folder and click “Refresh” to see your new files
- Open the folder called “lahars_raster”



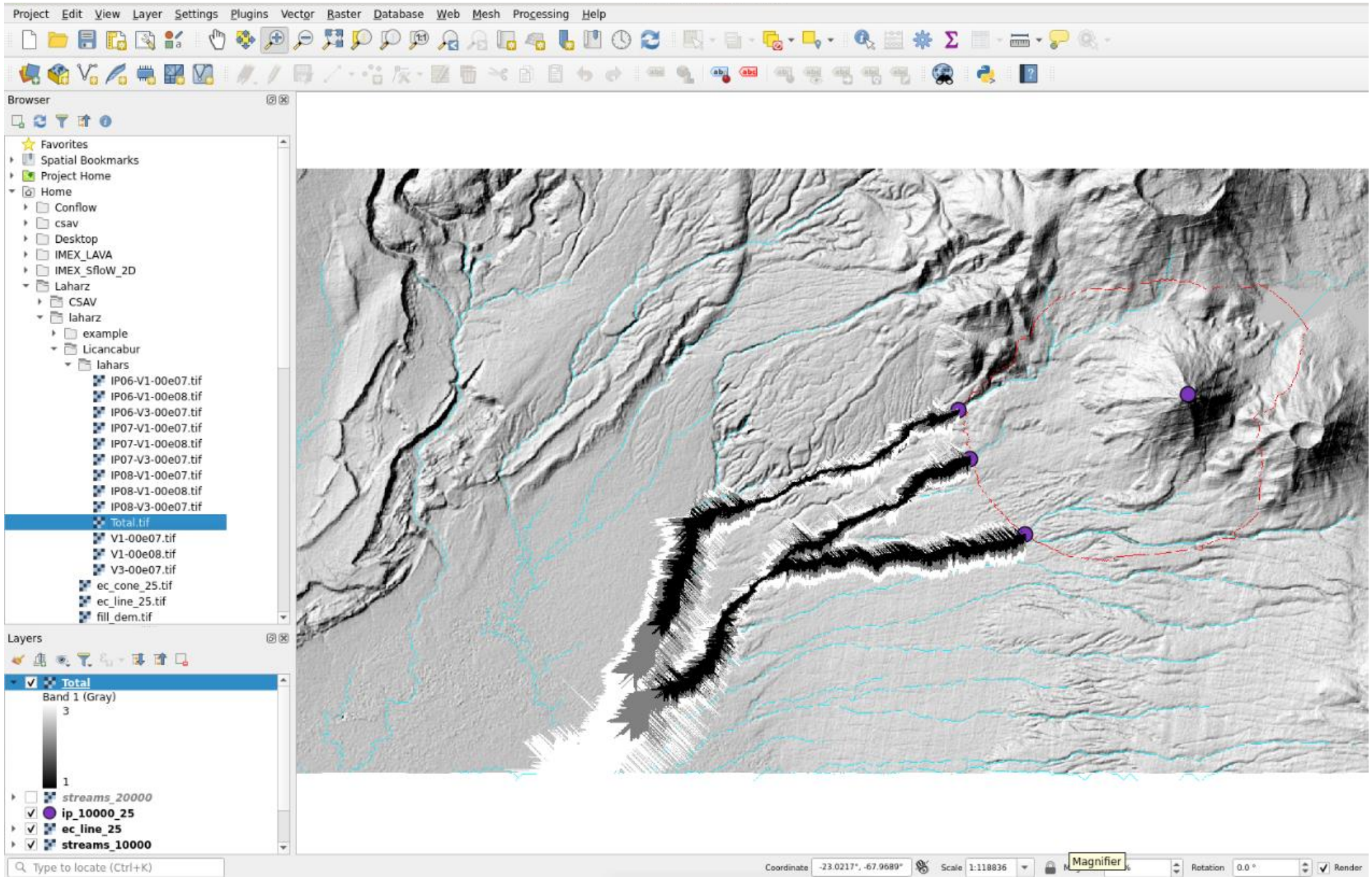


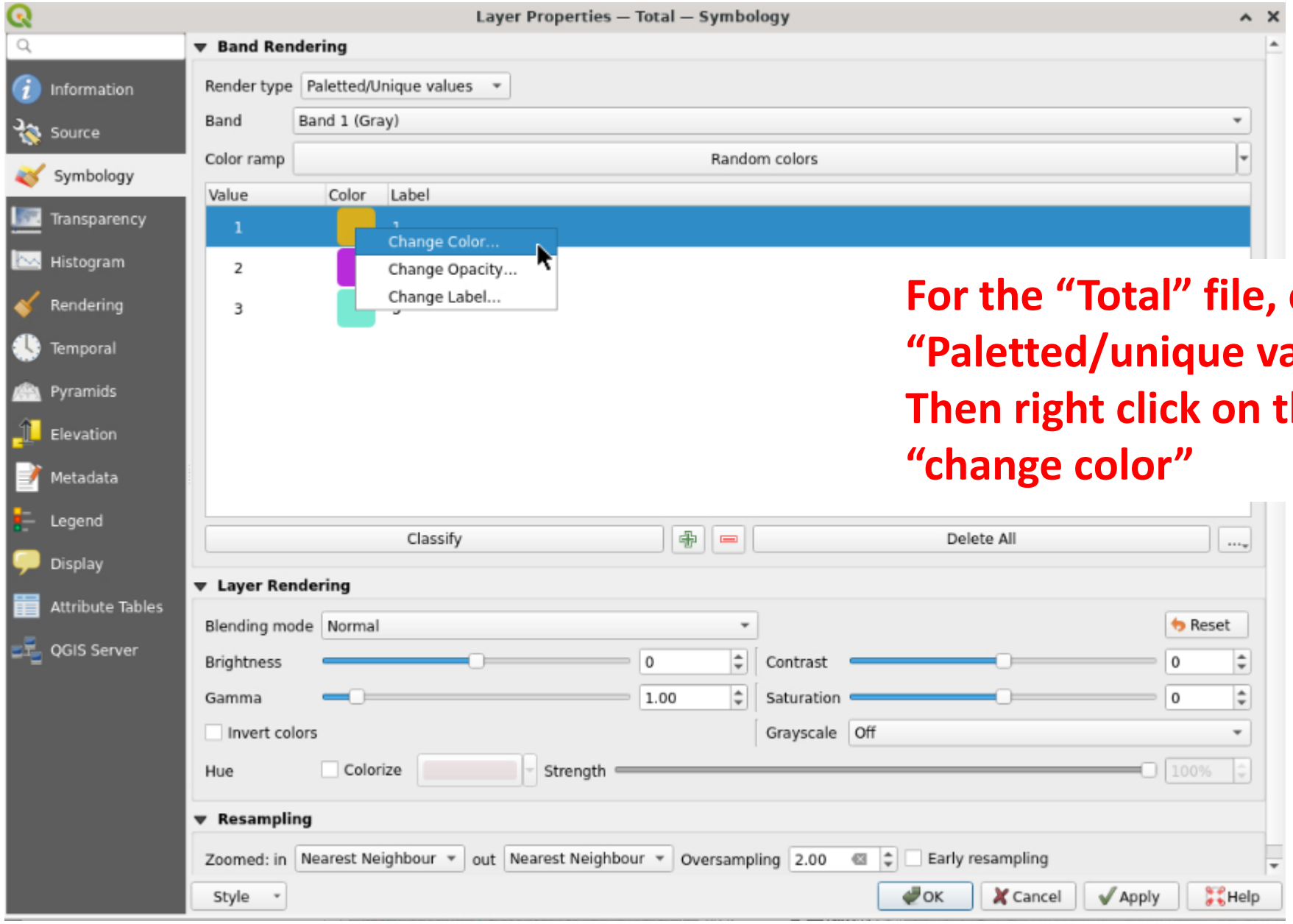
- LAHARZ made a bunch of different files
- There is a separate file for each volume/initiation point combination (these start with “IP”
- LAHARZ also made a file for each volume, merging all the flows from each initiation point (these start with “V”)
- And finally, there is a “Total” file that has all the volumes from all the initiation points in one file
- Add the “total.tif” file to your map



- LAHARZ made a bunch of different files
- There is a separate file for each volume/initiation point combination (these start with “IP”
- LAHARZ also made a file for each volume, merging all the flows from each initiation point (these start with “V”)
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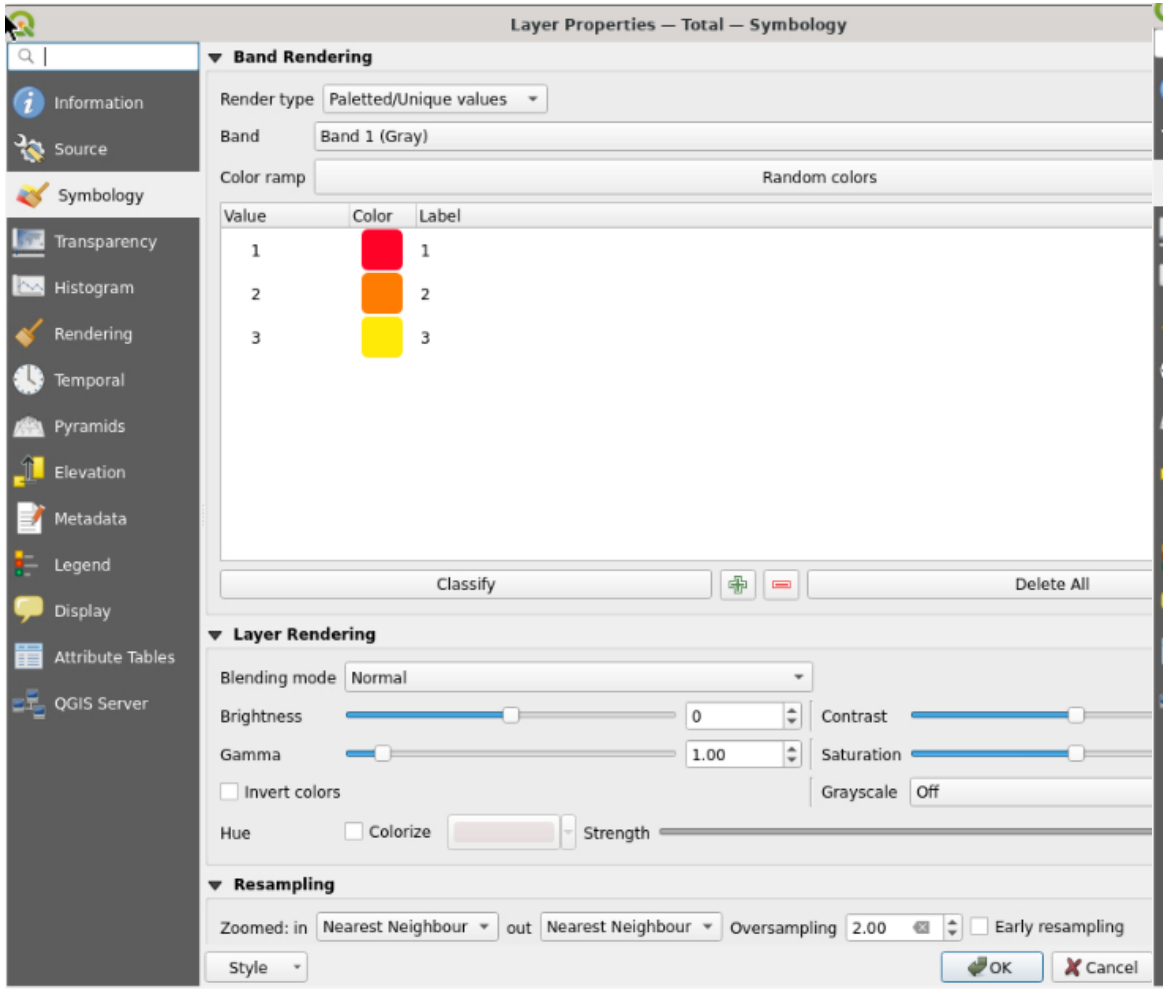
Change the colors and make the map prettier.

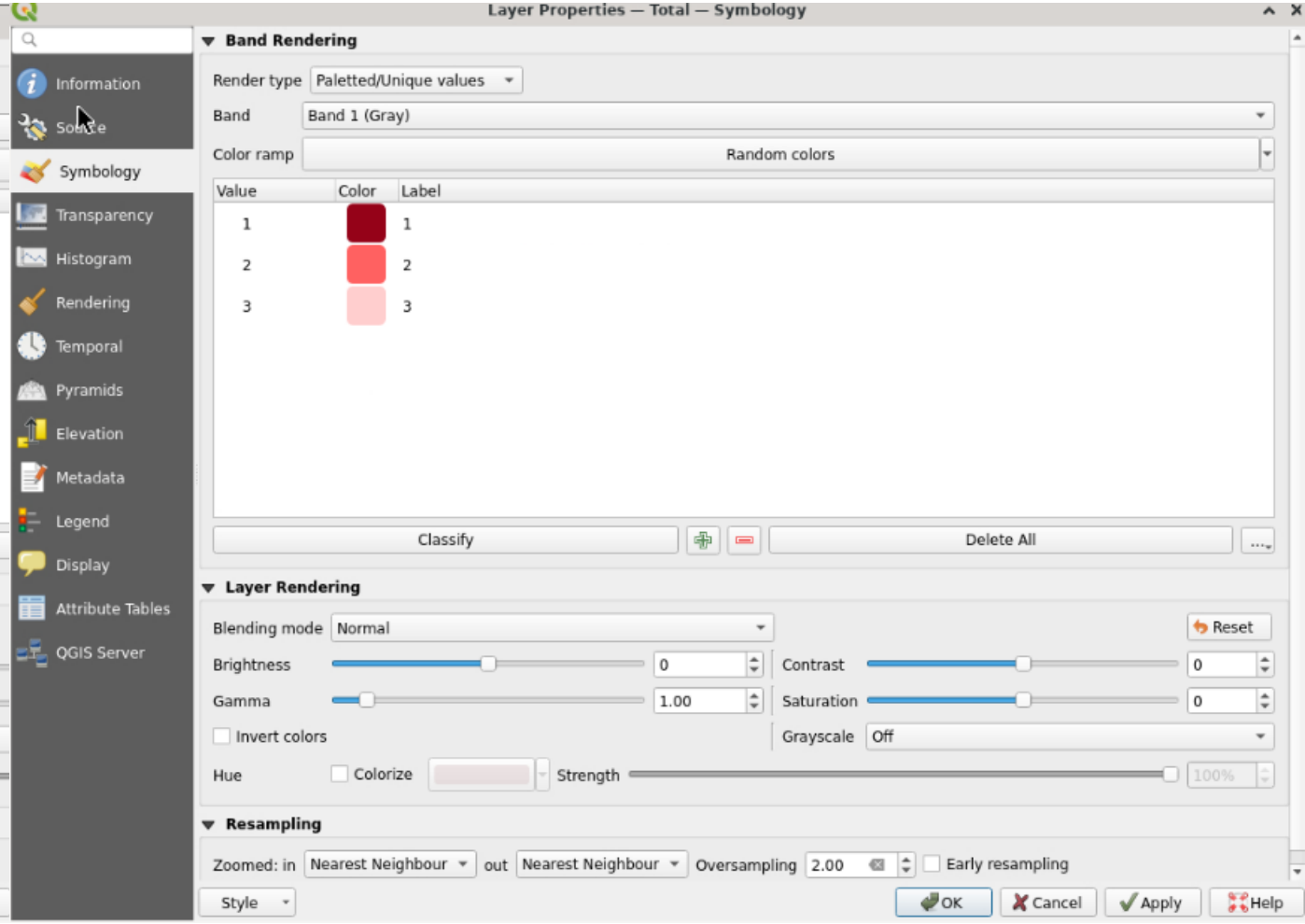


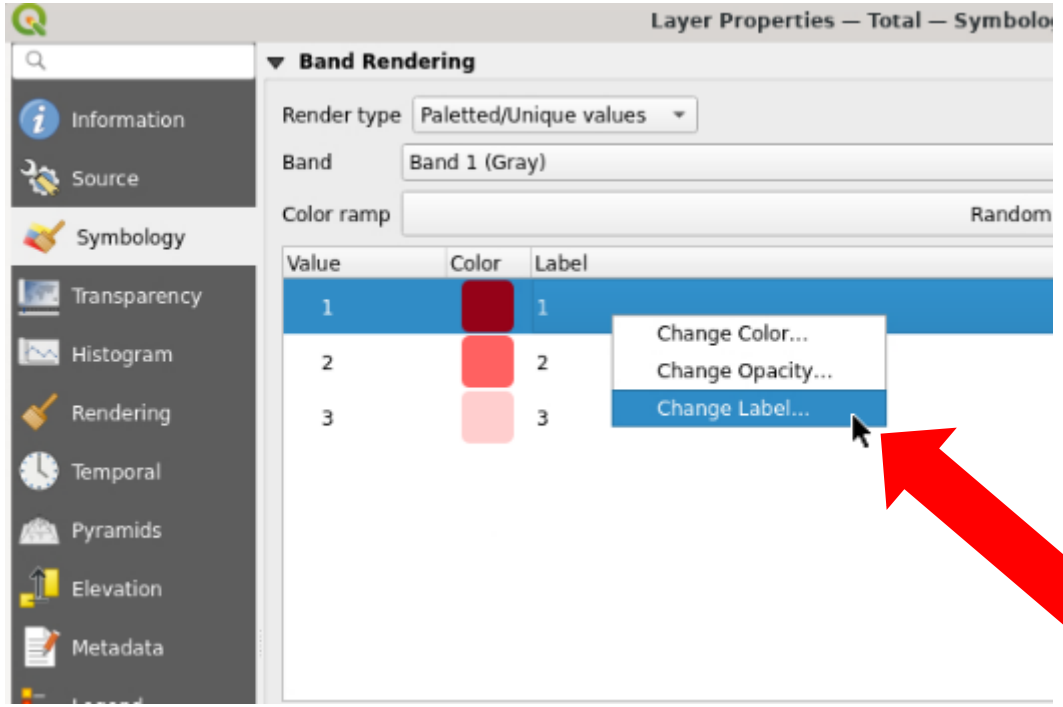


For the “Total” file, change the “render type” to “Paletted/unique values” and click “classify”. Then right click on the listed value and choose “change color”

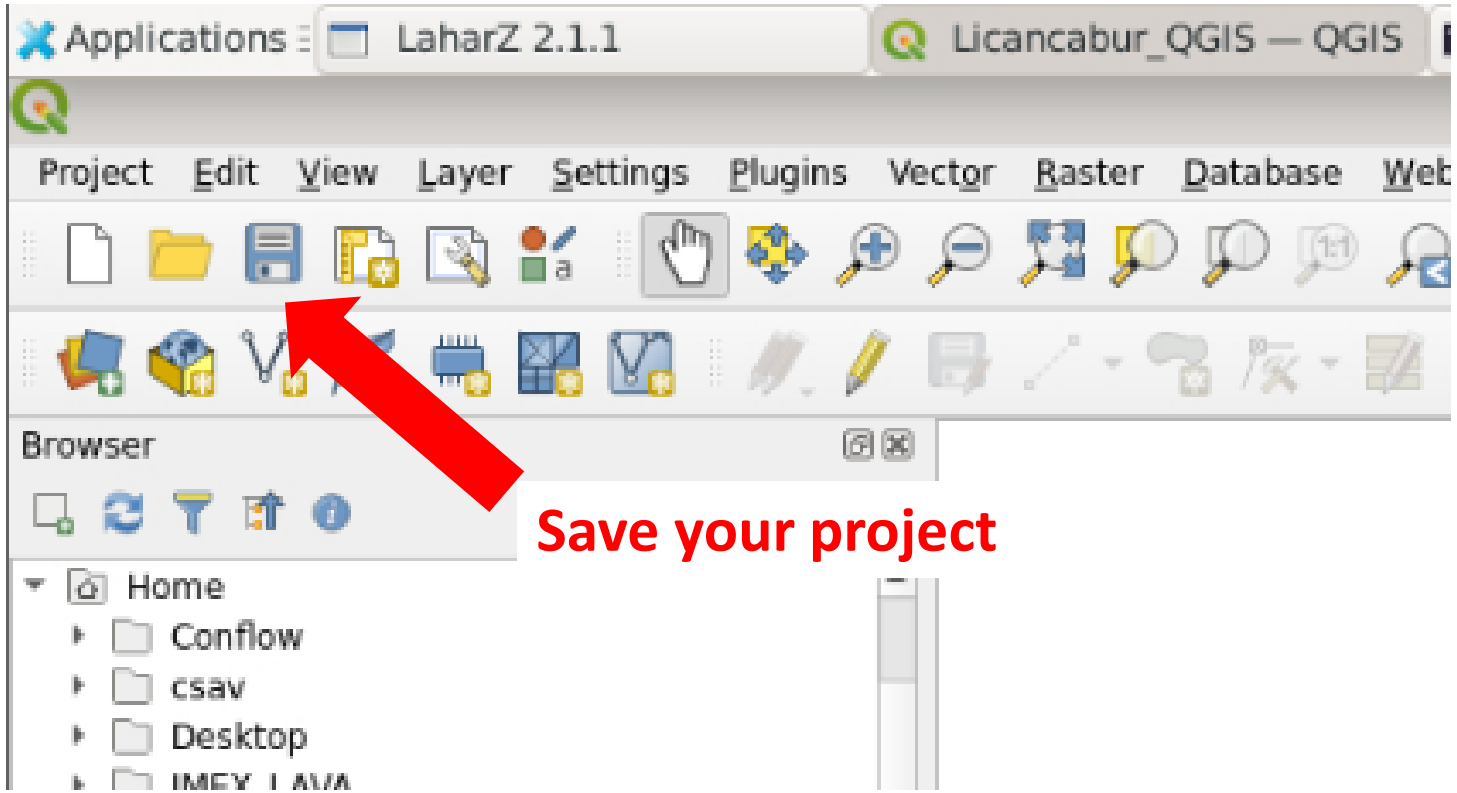
- The lowest value (1) is the lowest volume
- This is the highest hazard zone because smaller lahars occur more often.
- Choose a color scheme that makes sense, like red-orange-yellow or dark red to light red or anything else we discussed.





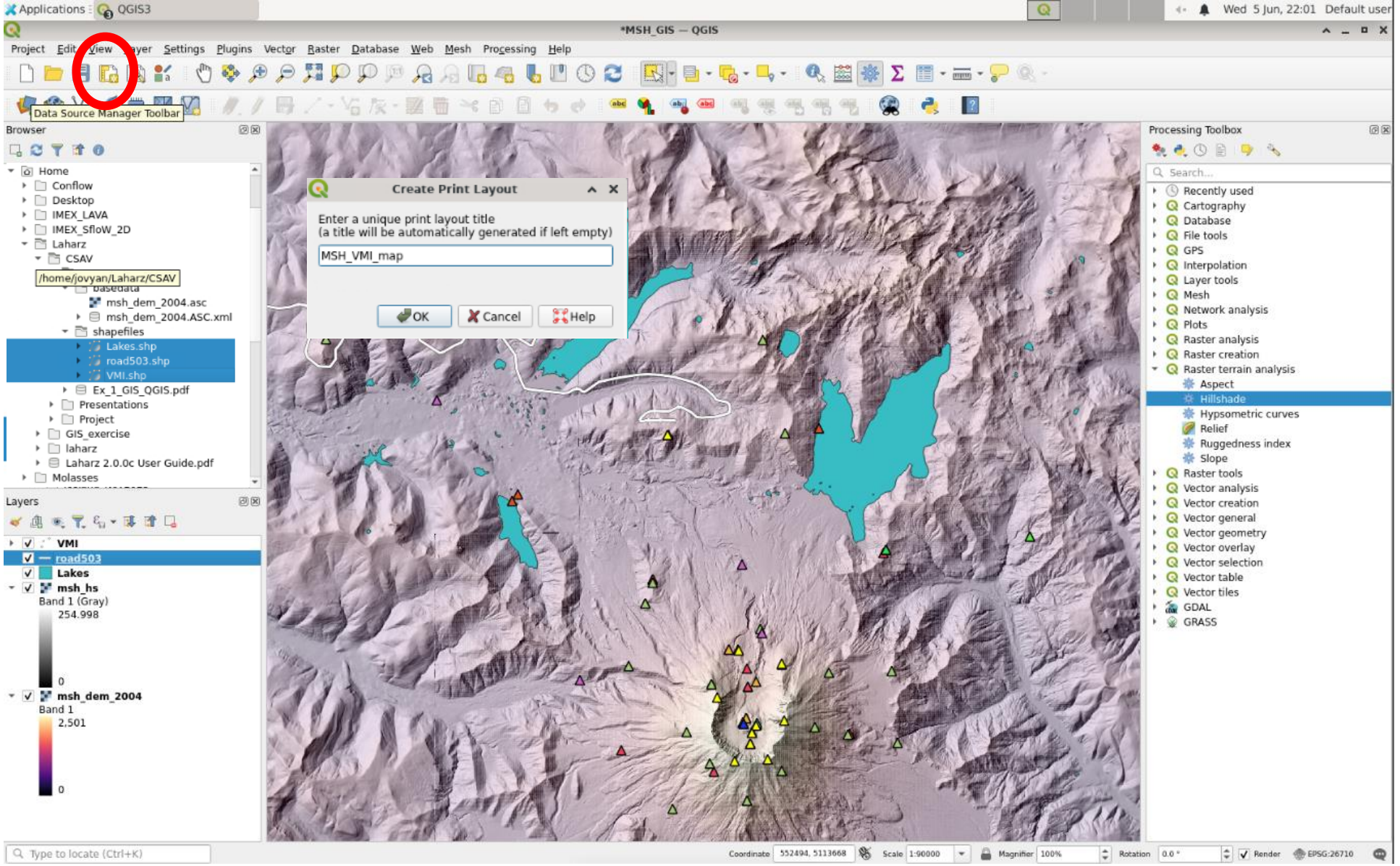


You can also change the label for each layer to the volumes you used

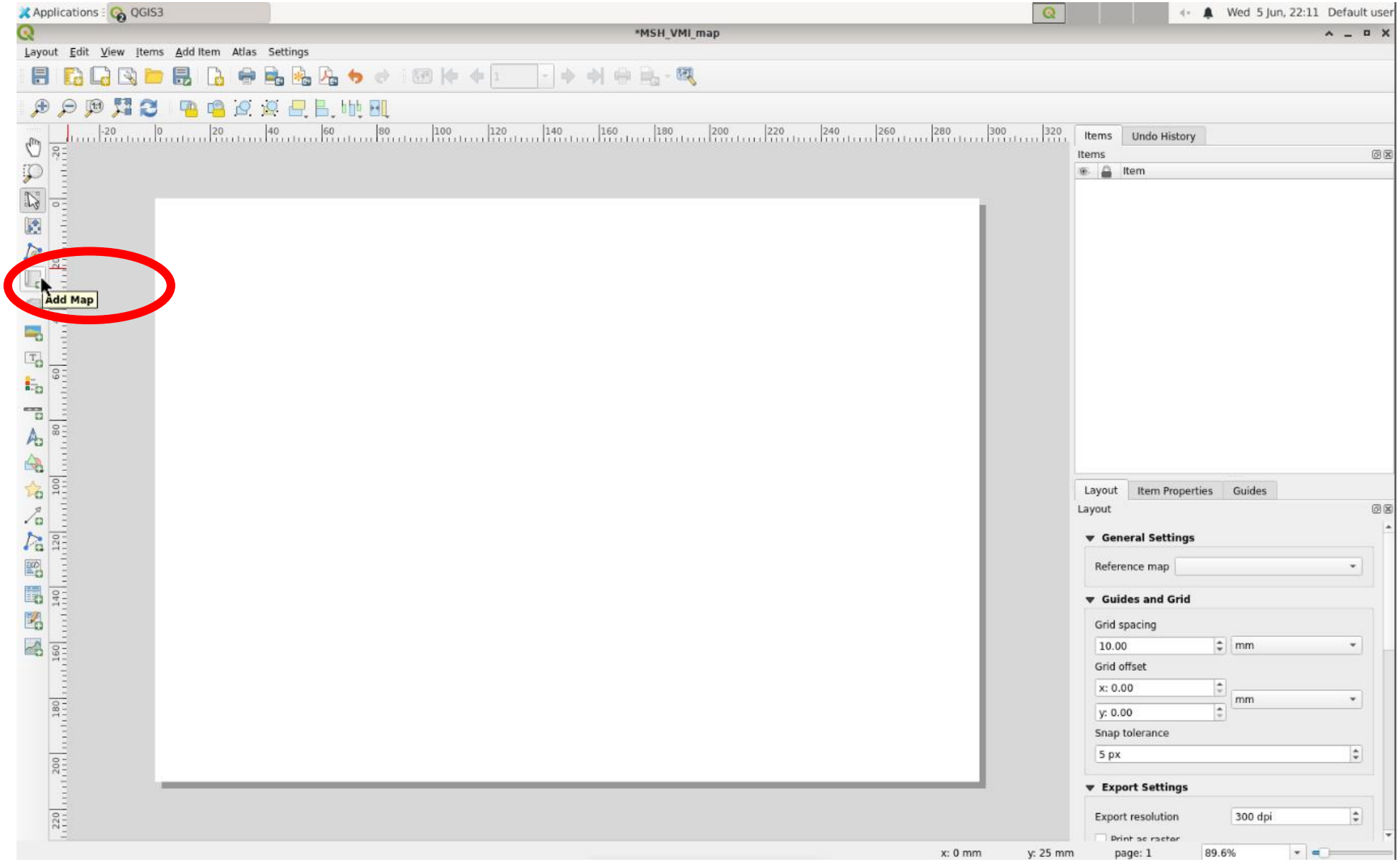

Applications
LaharZ 2.1.1
Licancabur_QGIS — QGIS
Project
Edit
View
Layer
Settings
Plugins
Vector
Raster
Database
Web
Browser
Save your project
Home
Conflow
csav
Desktop

Make a useful map!

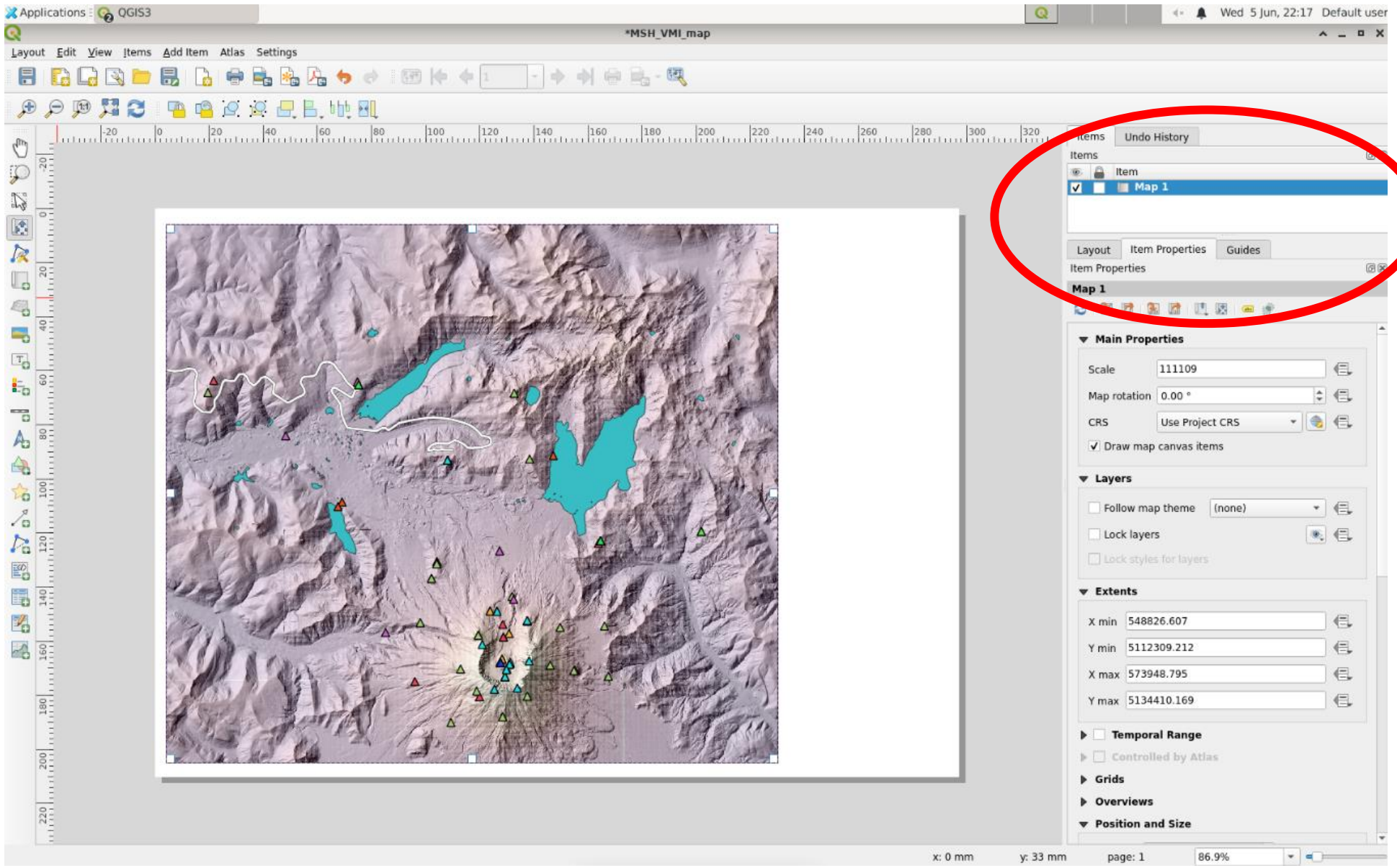
- Think about designing for your audience
- Refer to the “Ex_1_GIS_QGIS” exercise for instructions on:
  - Making the hillshade transparent and overlaying on the DEM
  - Making a map (“New Print Layout”)
  - Adding a title, legend, scale bar, north arrow, or any other elements you want to include!
  - I’ve included a few slides after this one from the GIS exercise



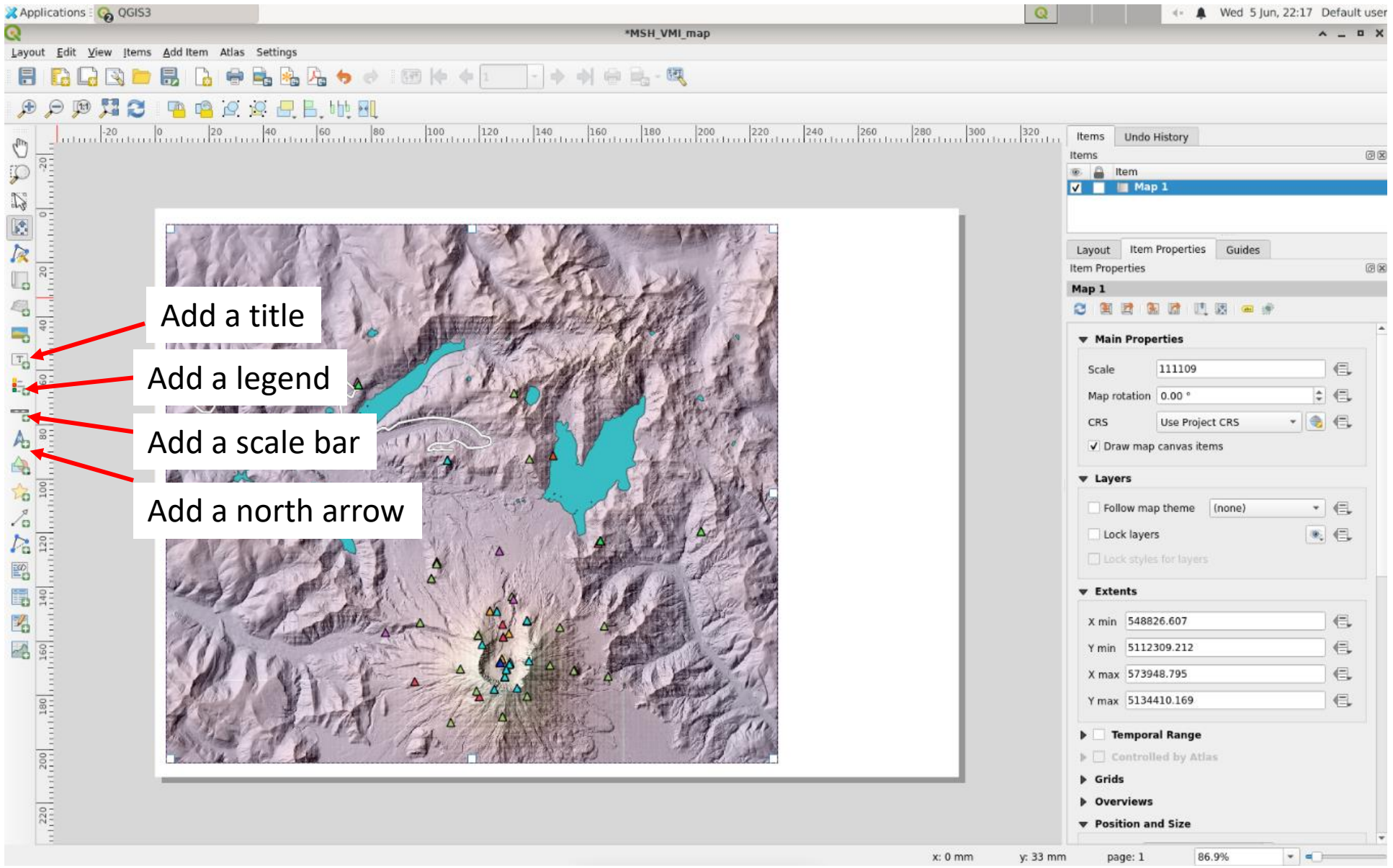
- Let’s make a map!
- Click the “New Print Layout” button
- Enter a name for the new map. “my_volcano_map”
- Click “OK”



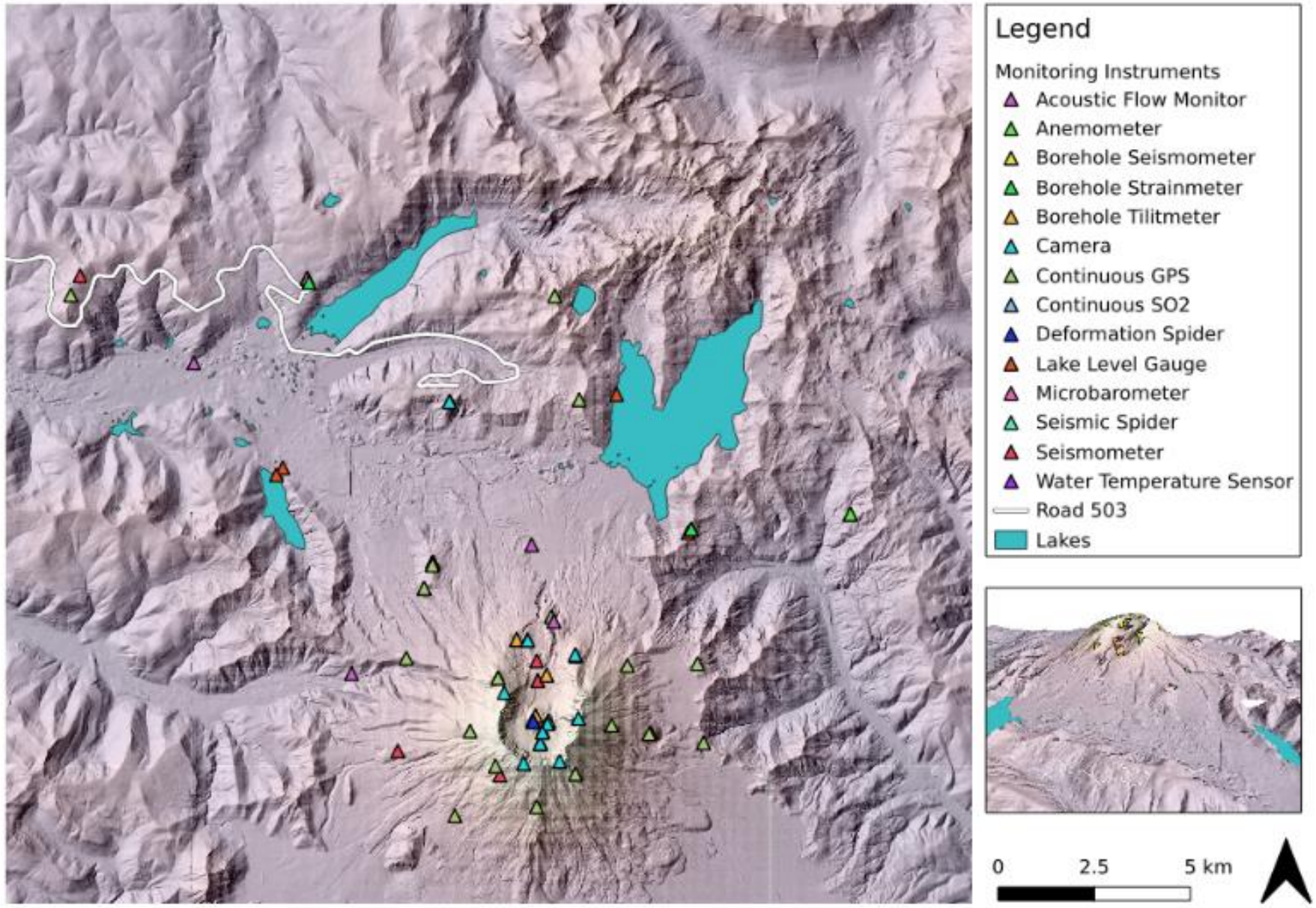
- First you need to add your map to the layout
- Click the “Add map” button
- Then draw a rectangle on the blank page to add the map



- You can adjust the scale and extent and other properties by selecting the map layer and “Item properties”



- You can add a legend for the stations and other layers
- Add a scale bar and/pr coordinates
- Add a north arrow
- Play around with the options to make your map



- Save your final work!
- You can also export the map as an image